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Experimental Investigation of Vortex-Induced Vibrations of Cantilevered Pipes: Modal Interactions and Effects of Aspirating Flow

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Abstract

Inspired by Seawater Intake Risers (SWIRs), an innovative experimental campaign tested, simultaneously, three vertical cantilevered flexible pipes fitted with a ballast attached at its tip in the scenarios of pure towing and towing with aspirating flow. Each model was designed to be excited by vortex-induced vibrations (VIV) in a different natural mode, namely the first, second and third modes. The corresponding natural frequencies associated with these modes were set to 1 Hz. An optical tracking system recorded the displacements of reflective tapes attached along the length of the models. The dynamic behavior of the flexible pipes was assessed by inspecting the displacement time series and the corresponding amplitude spectra along their lengths, as well as through modal amplitude time series and modal spectral content. Among the new aspects brought by the paper, we highlight experimental evidence of internal structural resonance effect, characterized by a transfer of energy from a natural mode in its upper branch of response to other modes. Additionally, the tests with combined conditions of towing and aspirating flow revealed that the influence of the aspirating flow on VIV is small, whether below or above the respective critical value. The results herein may serve as a new benchmark for mathematical models of cantilevered flexible pipes under the effects of pure VIV, as well as simultaneous VIV and aspirating flow.

Keywords: Cantilevered Flexible Pipe; Vortex-Induced Vibration; Aspirating Flow; Argand Diagram; Dynamics; Experiment

Nomenclature			
		m_H	Mass of the hose
		m_B	Mass of the ballast
ψ_n	Modal space function	m_M	Mass of the model
$A_{n\,x,y}$	Modal amplitude [mm]	m_{Miw}	Mass of internal water
$A^*_{n\,x,y} = A_{n\,x,y}/D$	Normalized modal amplitude	m_{Mdw}	Mass of displaced water
$\hat{A}_{n\,x,y}$	Modal amplitude spectra [mm]	m^*	Mass ratio
$\hat{A}^*_{n\,x,y} = \hat{A}_{n\,x,y}/D$	Normalized modal amplitude spectra	n	Natural mode of vibration
C	Structural damping coefficient	U	External flow velocity [m/s]
C_a	Added mass coefficient	U^*_y	Reduced velocity
D	External diameter [mm]	U^*_{ny}	Modal reduced velocity
D_i	Internal diameter [mm]	V_C	Aspirating flow critical velocity [m/s]
E	Young's modulus [N/m^2]	x, y	Coordinate [mm]
EA	Axial stiffness [N]	$x^*, y^* = x, y/D$	Dimensionless coordinate
EI	Bending stiffness [Nm^2]	X, Y	Amplitude spectra [mm]
$F_{x,y}$	Force component [N]	$X^*, Y^* = X, Y/D$	Dimensionless amplitude spectra
$f_{n\,x,y}$	Modal natural frequency [Hz]	z	Vertical coordinate [mm]
$F_{n\,x,y}$	Dominant modal frequency [Hz]	$z^* = z/L$	Dimensionless vertical coordinate
$F^*_{n\,x,y} = F_{n\,x,y}/f_{n\,x,y}$	Dimensionless dominant modal frequency	**Subscripts**	
I	Moment of inertia [m^4]	n	Natural mode of vibration
L	Length of the pipe [mm]	x	Streamwise direction
L_{ef}	Effective length of the pipe [mm]	y	Crosswise direction

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1. Introduction

In response to global efforts aiming at reducing greenhouse gas emissions, the offshore oil and gas industry has implemented measures over the past decades to reduce its carbon footprint. One such measure is the implementation of Seawater Intake Risers (SWIRs) in the operations of Floating Liquefied Natural Gas and Floating Production Storage and Offloading vessels. A Seawater Intake Riser is a vertical flexible pipe with a ballast attached at its free end, fixed at the hull for the purpose of aspirating cold water from depths that go beyond hundreds of meters.

The ocean environment exerts a complex influence on the dynamics of a seawater intake riser. The dynamic response is a consequence of loads resulting from the motions of the vessel and from the hydrodynamic forces associated to the internal and external water flows. This scenario may trigger complex responses, in which flow-induced vibrations and parametric excitation coexist.

In this context, the use of coupled computational fluid dynamics (CFD) and finite element analysis (FEA) simulations of SWIRs is practically unachievable due to the necessary numerical discretization (i.e., size of the meshes) in both the material and fluid domains. Notice also that, even when numerical results can be obtained, small-scaled experiments are of paramount importance since they provide data for benchmarking.

Regarding internal flow's effects, the foundation of the literature on the instability of cantilevered flexible pipes conveying fluid has started in 1961 by Benjamin with his theoretical and experimental studies of a system of articulated pipes ([4] and [5], respectively). Later on, Gregory and Païdoussis (1966, [18] and [19]) extended these studies to continuous flexible systems constrained to the horizontal plane (i.e. neglecting gravity effects) in which the behavior of the complex frequency was represented through a stability curve as a continuous function of the dimensionless internal flow velocity, unveiling the conditions of neutral stability and the transition from stable to unstable regimes for the lowest four natural modes of vibration of the system. As Païdoussis (1999, [31]) pointed out, the effect of the flow field at the outlet of the pipe is markedly different in ejecting and aspirating conditions. In the former, the flow field generates a jet that can be interpreted as a follower force, whereas in the latter, a sink-like flow field is formed. The theory of pipes aspirating fluid indicates that weak instability occurs at infinitesimal flow velocities. However, the effects of external fluid damping make experimental confirmation challenging. Consequently, CFD simulations must be employed to elucidate this phenomenon (Païdoussis, 2005 [32]). More recently, Chen *et al*. (2019, [8]) presented a geometrically exact equation for large amplitude oscillation for a flexible cantilevered pipe discharging fluid, accounting for the full geometric nonlinearities due to curvature effects, recovering limit-cycle oscillation after flutter instability. Adiputra and Utsunomiya (2019, [1]), compared an analytical solution to numerical analysis of a cold-water pipe system in aspirating condition in order to determine the best top joint connection, plastic material and necessity of ballast use.

A particular FIV phenomenon caused by the external flow is known as vortex-induced vibrations (VIV), Feng (1968, [14]), Bearman (1984, [3]), Blevins (2001, [7]), Sarpkaya (2004, [37]), Bernitsas *et al*. (2019, [6]). Experimental studies on the modal responses of cantilevered flexible cylinders under pure VIV excitation were undertaken by a variety of authors. These studies have reinforced the intrinsic nonlinear nature of such phenomenon. In Pesce and Fujarra (2000, [34], and 2005, [35]), the authors presented evidences of a jump phenomenon and the existence of an upper overlapping branch of response at the range $8 \leq U^*_{1y} \leq 10$, with amplitudes larger than those presented by the upper branches of rigid cylinders. In Fujarra *et al*. (2001, [17]), a stable branch of response coexisting with the usual desynchronization region was reported in tests conducted with a flexible cantilevered cylinder with orthotropic bending stiffness. Such a distinct branch of response was named high-amplitude branch, for exhibiting amplitudes larger than those presented in the respective upper branch. Defensor Filho, Franzini and Pesce (2022, [12]) discussed the mechanism of extending the region of resonance to higher reduced velocities by controlling the in-line to crosswise natural frequencies ratio of a cantilevered flexible cylinder, thus giving rise to a multimodal response. In Bai *et al*. (2024, [1]), the authors obtained vortex-induced force coefficients through a hydrodynamic identification method. Nonlinear characteristics in larger flow velocities were pointed out, such as unstable VIV amplitudes, which resulted in coefficients that exhibited time-varying features. Additionally, a complex in-line frequency response was observed. The dominant peak presented an in-line to crosswise frequency ratio equal to 1.0 (a fact that was attributed to the inclination of the free end due to drag effects), alongside wide-banded harmonic components with twice the value of the dominant crosswise frequency.

The aforementioned examples highlight quite different dynamic features in comparison to the classical VIV problem of a rigid cylinder mounted in elastic base, adding more complexity to the pursuit of a physical-mathematical model that can reliably represent such fluid-structure interaction regardless of the structural configuration.

The complexity is further increased in the case of the derivation of an analytical model aiming at reproducing the dynamics of a SWIR. Ensuring consistency in the physical-mathematical modeling of the concomitant external and internal flows, which are dependent on the kinematic state of the pipe, is a main challenge, rather than the structural modeling itself. Such a modelling complexity is reflected as scarcity of works in the literature addressing flexible pipes under the concomitant effects of both the external and internal flows. However, as the topic has gained prominence in the ocean engineering field, focus has been placed on it, resulting in a sensible increase in research efforts dedicated to its study.

In this scenario, recent theoretical studies examining the effects of internal flow conveyance on the VIV responses of flexible pipes were conducted by many authors. Dai *et al*. (2014a, [9]) investigated a vertically supported flexible pipe concomitantly subjected to uniform external flow, internal fluid conveyance, and harmonic upper base horizontal excitation. The findings show that the VIV amplitude can be increased or decreased in combination with the base excitation, as well as a wider VIV synchronization region may result as an increase of the base acceleration. In Dai *et al*. (2014b, [10]), the effect of pulsating internal flow, which is assumed to have a harmonically varying component superposed on a steady mean velocity, is investigated on the external cross flow response of a two-dimensional theoretical model of a vertical supported flexible pipe. Among the findings is that the peak of VIV amplitude is larger with pulsating internal flow than with steady flow. Meng, Kajiwara and Zhang (2017, [21]) evaluated the effects of discharging flow, both below and above the respective critical velocity, on a two-dimensional cantilevered flexible pipe subjected to VIV. The findings pointed out that the internal flow can vary the riser natural frequencies, thus triggering new modes and switching the dominant mode. Additionally, the internal flow velocity below the respective critical value damps the VIV responses while when it is above the critical value the VIV response is enhanced, moreover, the internal flow critical velocity is increased with the increase of the external current velocity. Orsino *et al*. ([22] to [30]) presented various studies with a 3D nonlinear reduced-order model of a cantilevered flexible pipe, examining the effects caused by aspirating and discharging flows below and above their respective critical velocities, under the concomitant VIV excitation. Duan *et al*. (2021, [13]), in an investigation with a finite element model of a vertically supported flexible pipe, examined the effects of aspirating flow with varying velocities and densities on the VIV response caused by external shear flow. The authors pointed out that multi-frequency response is induced when both excitations are concomitant, increased internal flow velocity and densities enlarge the in-line mean deflection, and different internal flow velocities and densities cause mode and frequency transitions responses. Zhao *et al*. (2023, [38]) examined the effects of aspirating flow with different densities on a uniform external flow VIV response of a vertically supported flexible pipe. Their findings corroborated the hypothesis that the aspirating flow has a non-negligible effect on the IL dynamic response. Additionally, the travelling wave characteristics of the system are able to be enhanced by the internal flow, and that the transmission direction of the travelling wave is the same as the direction of internal flow velocity.

In this context, to the best of the authors' knowledge, no studies have been conducted to date that address, experimentally, cantilevered flexible pipes with internal flow in aspirating condition concomitantly subjected to VIV. The reason behind this interest lies in understanding how, if at all, the aspirating flow affects VIV. Therefore, we conducted an experimental campaign to address this gap and to calibrate an in-house numerical code used in simulations of SWIR systems. The objective of the tests was to assess the dynamics of a fundamental model of a seawater intake riser, rather than a scaled one, with the aim of obtaining conceptual understanding rather than scaled information. Two main scenarios were investigated: (i) pure VIV, and (ii) combined VIV and aspirating flow. The latter scenario, as aforementioned, was planned with the objective of revealing mutual influences of both phenomena and determining which one has dominant effects. Additionally, the present work aims to provide a benchmark for analytical and numerical models elaborated to deal with this kind of dynamical problem.

The present text is structured as follows: after this introduction, Section 2 brings the experimental methodology detailing the experimental setup and the analysis methodology. The results are presented in Section 3, and the final conclusions are addressed in Section 4. Appendix A provides additional information on the design and construction of the experimental models.

2. Experimental Methodology

Based on the assumption that the investigation aimed to study the fundamental dynamics of a SWIR, the experimental models were designed and constructed based on the following conceptual features: (i) a flexible pipe with a rigid ballast attached to the tip; (ii) both the flexible pipe and the ballast would have the same internal and external diameters; and (iii) the system would be mounted in a vertical cantilevered configuration.

In order to obtain a richer data basis for response correlations, three different models were designed with the same pipe and ballast diameters but different lengths, therefore resulting in different natural modes excited at the same external flow velocity by VIV. The first, second and third natural modes of vibration were chosen to be the dominant modes with the three corresponding natural frequencies settled to be as close as possible to 1 Hz ($f_{1y} \approx f_{2y} \approx f_{3y} \approx 1.0$ Hz). The models were identified as BH-1, BH-2, and BH-3, where the acronym BH refers to the ballasted-hose concept, and the sequential numbers indicate the natural mode planned to be dominantly excited by VIV.

The flexible pipes were made from a rubber hose with an internal fiber reinforcement layer, whose material presented the measured linear density $\mu_H = 0.56$ kg/m, equivalent dynamic bending stiffness $EI_H = 1.23$ Nm^2, and equivalent dynamic axial stiffness $EA_H = 12.5 \times 10^3$ N. The complete description of the rubber hose characterization procedure is presented in the Appendix A. Brass was selected as the ballast's material due to its high density and good machinability, which presented, after manufacturing, linear density $\mu_H = 4.15$ kg/m. Due to its higher stiffness compared to the hose's, the brass theoretical Young's modulus $E_B = 105 \times 10^9$ N/m^2 was considered for calculating the associated axial and bending stiffnesses values, $EA_B = 1.25 \times 10^4$ N and $EI_B = 4.91 \times 10^3$ Nm^2, respectively. Both the hose and the ballast present internal and external diameters of $D_i = 22$ mm and $D = 33$ mm, respectively.

Based on the material parameters and the planned natural frequencies, the lengths of the hoses and ballasts were calculated using an implementation of the 3D reduced-order cantilevered flexible pipe physical-mathematical model, whose derivation and applications can be found in [22] to [30]. Hereinafter, this mathematical model will be referred to as 'in-house' code.

The experimental models were manufactured as accurately as possible according to the specified lengths. The as-built total lengths, which is the sum of the flexible pipe and ballast lengths, were measured and found to be 655 mm, 1705 mm and 3030 mm, for models BH-1, BH-2, and BH-3, respectively. The natural frequencies in water were evaluated using the in-house code settled with the as-built experimental lengths, which are depicted in Tab. 1 along with the measured values from water decay tests and the respective differences between then. It is important to note that, since the rubber hose was commercially acquired rather than manufactured, it exhibits some residual curvature due to its coiled storage following the manufacturing process. This imperfection was not considered in the in-house numerical code, which assumed a perfectly vertical cantilevered pipe. The residual curvature causes slight discrepancies in the natural frequencies between the orthogonal measurement directions, contrary to what is considered in the numerical model. Nevertheless, it is notable that the discrepancies remain within an acceptable range. In the experimental setup, attention was paid to prevent undesired static equilibrium configurations outside the central longitudinal drift plane. This was achieved by orienting the curvature concavity to the downstream flow direction.

Table 2 presents the nominal aspirating flow critical velocities which were calculated with *root-loci* diagrams considering the models submerged, in the trivial equilibrium configuration, and in the absence of external flow. Figure 1 shows the respective *root-loci* diagrams. The aspirating critical velocity values were obtained as the minimum one associated with positive value of the real part of the eigenvalues computed from the Jacobian Matrix (Lyapunov's indirect Method) and used as reference values in the experiments (as further explained in section 2.1). The *root-loci* diagrams display the natural periods of the experimental models ordered from top to bottom, as a function of the internal flow velocity (horizontal axis). A color scheme is used to indicate the real part of the eigenvalues. Blue and red colors denote stability and instability, respectively, of the trivial equilibrium configuration. Flow velocities in aspirating condition are shown on the positive side of the horizontal axis. For completeness, an equivalent section of the negative horizontal axis indicating flow velocities in discharging condition is also included. The vertical dashed black lines indicate the velocities at which the eigenvalues' real parts shift from negative to positive values, indicating the threshold from stable to unstable regimes, and defining the internal critical flow velocities. All three *root-loci* exhibit the aspirating threshold of instability in the first natural mode of vibration at marginal velocities (as previously reported; [31], [32]): 0.06 m/s for model BH-1; 0.03 m/s and 0.02 m/s for models BH-2 and BH-3, respectively. Table 3 summarizes the parameters of the experimental models.

Table 1: Calculated and measured as-built natural frequencies in water and the respective differences (in the parenthesis) to the calculated as-built values.

Experimental model	Direction	Calculated natural frequency in water f_n [Hz]			Measured natural frequency in water f_n [Hz]		
		$n = 1$	$n = 2$	$n = 3$	$n = 1$	$n = 2$	$n = 3$
BH-1	x	0.98			1.03 (5.43 %)		
	y	0.98			1.04 (6.46 %)		
BH-2	x	0.28	1.12		0.29 (4.05 %)	1.15 (2.96 %)	
	y	0.28	1.12		0.27 (3.12 %)	1.18 (5.52 %)	
BH-3	x	0.17	0.56	1.14	0.18 (4.08 %)	0.56 (0.31 %)	1.17 (2.53 %)
	y	0.17	0.56	1.14	0.17 (1.90 %)	0.56 (1.17 %)	1.15 (0.87 %)

Table 2: Aspirating flow critical velocities. Evaluation from the in-house code.

Experimental model	Mode that becomes unstable	Critical flow velocity V_C [m/s]
BH-1	1	0.06
BH-2	1	0.03
BH-3	1	0.02

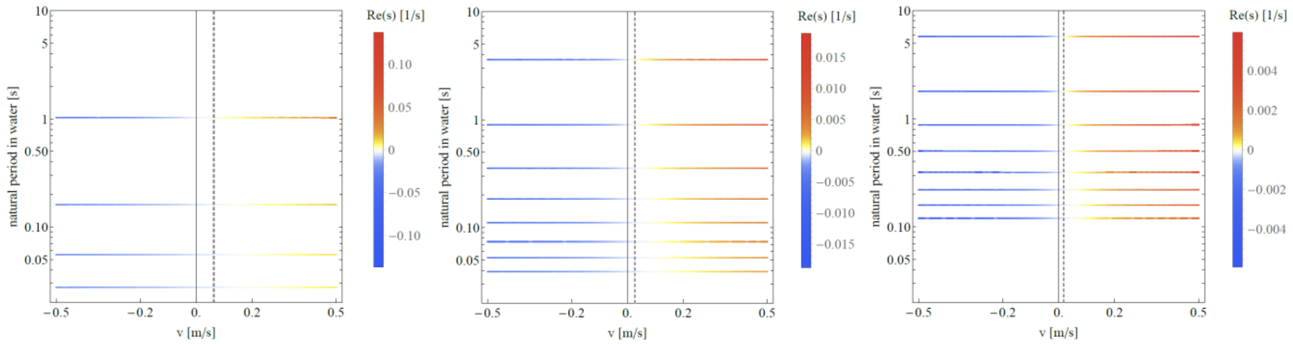


Figure 1: *Root-loci* diagrams for the experimental models submerged in water, in the trivial equilibrium configuration, and without external flow. From left to right: experimental model BH-1, BH-2 and BH-3. Positive velocity values indicate aspirating cases.

Table 3: Experimental models parameters.

	Parameter	BH-1	BH-2	BH-3
Geometric	External diameter - D [mm]	33	33	33
	Internal diameter - D_i [mm]	22	22	22
Hose	Material	Nitrile rubber	Nitrile rubber	Nitrile rubber
	Young's modulus - E_H - MPa (10^6) [N/m^2]	26.38	26.38	26.38
	Axial stiffness - EA_H [N]	1.25E+04	1.25E+04	1.25E+04
	Bending stiffness - EI_H [Nm^2]	1.23	1.23	1.23
	Linear density - μ_H [kg/m]	0.56	0.56	0.56
	Length - L_H [m]	0.595	1.595	2.857
	Mass - m_H [kg]	0.33	0.89	1.60
Ballast	Material	Brass	Brass	Brass
	Young's modulus - E_B - MPa (10^6) [N/m^2]	105.0E+3	105.0E+3	105.0E+3
	Axial stiffness - EA_B [N]	4.99E+07	4.99E+07	4.99E+07
	Bending stiffness - EI_B [Nm^2]	4.91E+03	4.91E+03	4.91E+03
	Linear density - μ_B [kg/m]	4.15	4.15	4.15
	Length - L_B [m]	0.060	0.110	0.173
	Mass - m_B [kg]	0.25	0.46	0.72
Complete model	VIV dominant natural mode - n	1	2	3
	Measured natural frequency of the mode n in water - f_{ny} [Hz]	1.04	1.18	1.15
	Length ratio - L_B/L_H [%]	10.16 %	6.90 %	6.06 %
	Total length - L_M [m]	0.655	1.705	3.030
	Slenderness - L_M/D	19.85	51.67	91.82
	Mass - m [kg]	0.58	1.35	2.32
	Mass of internal water - m_{iw} [kg]	0.25	0.65	1.15
	Mass of displaced water - m_{dw} [kg]	0.56	1.46	2.59
	Mass ratio - m^*	1.49	1.37	1.34
	Aspirating critical flow velocity in water - V_C [m/s]	0.06	0.03	0.02
	Unstable mode	1	1	1

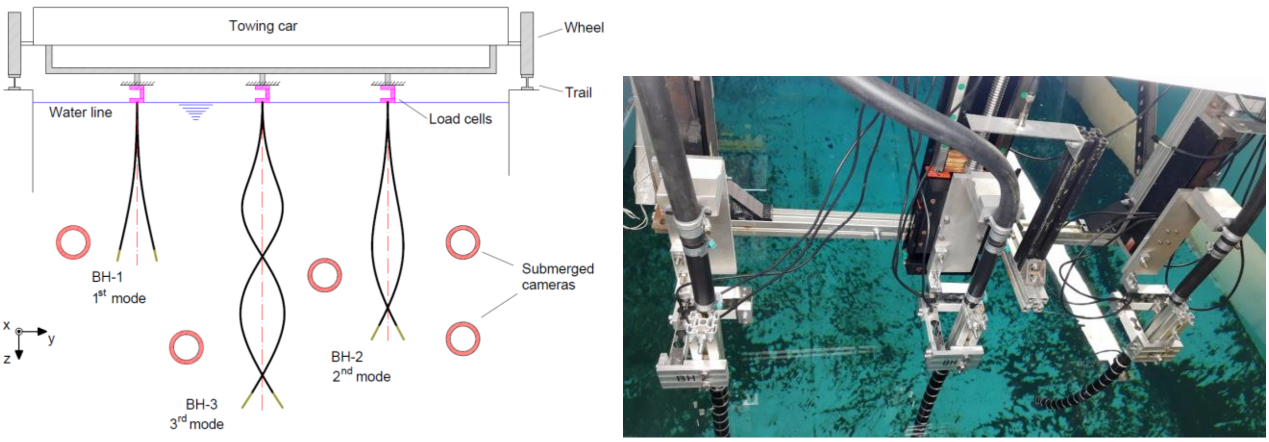
2.1. Experimental setup

The experiments were conducted at the towing tank facility of the Technological Research Institute of the State of São Paulo (IPT). The tank is 240 m long, 6 m wide and 4 m deep, with the motion provided by a rail-oriented electric towing car. X, Y and Z were adopted for the in-line, transverse and vertical directions with respect to the incoming flow, respectively.

A pumping system based on the calculated aspirating flow critical velocities was assembled on an auxiliary dispositive, fixed to the towing car, to ensure that no vibrations were transmitted to the experimental models. The aspirated water was returned to the tank downstream the experimental models through a pipe sufficiently distant to avoid any hydrodynamic interference. The same care was taken with the lateral distance between the three simultaneously towed models.

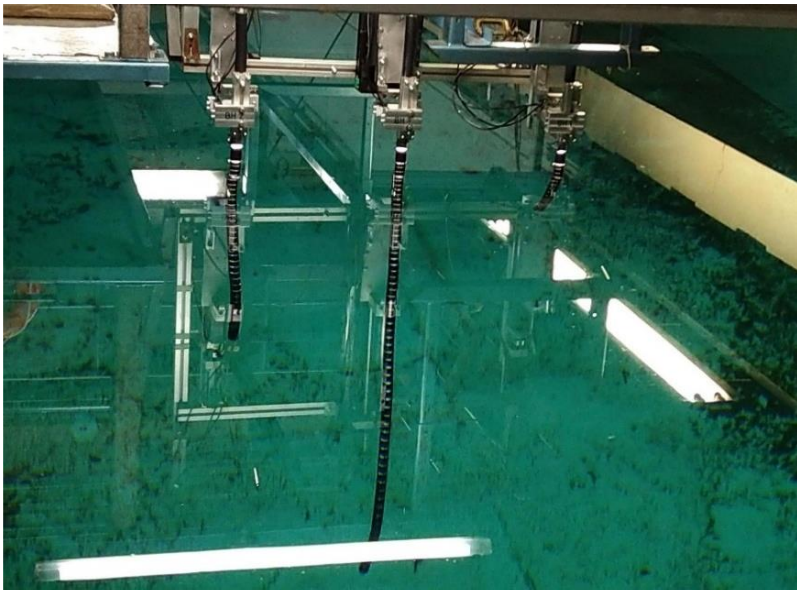
Three Alpha Instrumentos® GLX-15 uniaxial stainless steel load cells were assembled as a block force to measure the force components (F_x, F_y and F_z) at the models' clamps. The load cells nominal capacity is 150 N with a sensitivity of 0.1%. To acquire the displacements, an optical tracking system measured the 3D Cartesian coordinates of reflective tapes (or targets) placed along the length of the models. Model BH-1 used 8 reflective targets, while models BH-2 and BH-3 used 18 and 32 targets, respectively. The Lynx ADS 2000 system was used to acquire data from all instruments and equipment, with a sampling frequency of 30 Hz.

The campaign towed the experimental models in a side-by-side arrangement, ensuring that the same external flow velocity was imposed to each model in every test (see Figure 2). The lateral distance between the models was set as 500 mm, sufficient to avoid hydrodynamic interference. The maximum towing velocity was set at approximately 0.52 m/s to prevent excessive vibrations resulting from the rail-wheel contact, as well as free surface effects. Twenty towing speeds were selected to cover the range of external flow velocities investigated. As previously mentioned, the models were positioned with the concavities caused by residual curvatures oriented towards the opposite direction to the flow. This was done to prevent the formation of equilibrium positions that would deviate from the central vertical plane. For the scenario of simultaneous excitation by the internal and the external flows, five towing speeds were repeated with three different aspirating flow velocities based on the calculated critical values for each model. The three aspirating velocities considered, regarding the critical value (V_C) are: 0.5 V_C, 1.0 V_C, and 2.0 V_C.



(a) Representation of the experimental setup. From left to right: models BH-1, BH-3 and BH-2.

(b) Load cells and anchorage assemblage.



(c) Experimental models in test position. From left to right: BH-2, BH-3, and BH-1.

Figure 2: Photos of the experimental setup.

2.2. Data analysis

A methodology combining time-domain and frequency-domain techniques was developed for this investigation, detailed as a flowchart scheme in Figure 3. The results comprise (i) amplitude and frequency responses along the models' length, and (ii) modal responses. To illustrate the methodology, the steps are presented in the case of the experimental model BH-3 at the maximum reduced velocity tested, $U^*_{3y} = 13.71$ ($U = 0.52$ m/s), where $U^*_{ny} = U/f_{ny}D$.

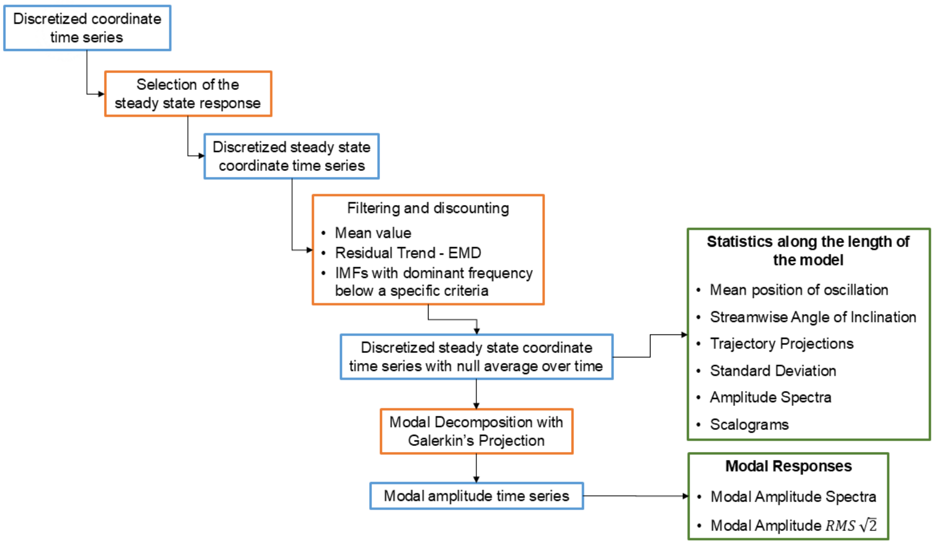


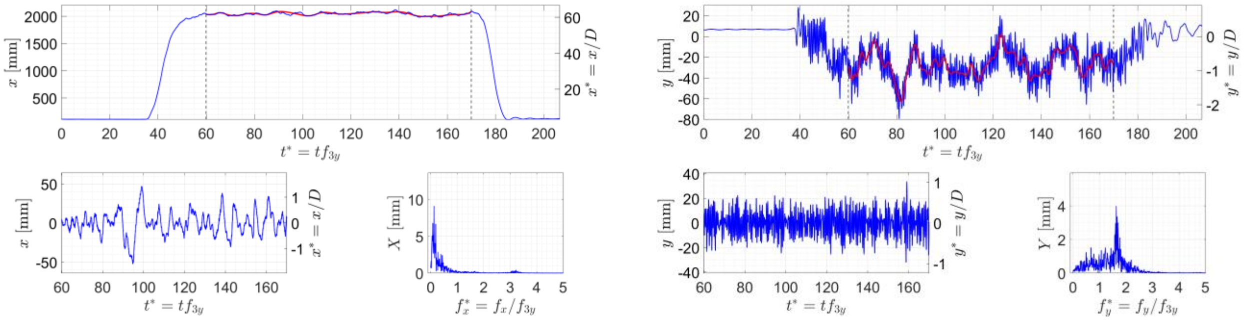
Figure 3: Data analysis flowchart. Blue frames indicate the type of signal, orange frames indicate the technique or procedure used, and green frames indicate the results obtained.

Initially, the steady-state regime of the time series is selected, thus excluding the initial and final transients associated with the increase and decrease in towing speeds. Subplots Figure 4(a) and Figure 4(b) illustrate it using the streamwise and crosswise displacements measured at the tip of model. The black vertical dashed lines indicate the selected portion of the signal.

To ensure null average over time in the selected portion of the signal, the respective mean value, residual trend, and low-frequency content are filtered out. The residual trend and the low-frequency content are assessed through the Empirical Mode Decomposition (EMD) method, which constitutes the initial part of the Hilbert-Huang Transform technique (HHT), a highly effective tool for dealing with non-stationary signals. See Huang *et al.* (1998, [20]) for the technique, and Pesce *et al.* (2006, [33]) and Defensor Filho *et al.* (2022, [12]) for applications of the technique in VIV problems. The EMD procedure sifts a signal into a set of time series named Intrinsic Mode Functions (IMFs), each one representing a distinct oscillation frequency embedded into the signal. The instantaneous average, or residual trend, is the last information provided along with the IMFs. The low-frequency content was obtained with an *ad-hoc* criteria based on the frequencies of interest in the models, which identify and discount from the signal the IMFs with the dominant amplitude peaking marginal frequencies (close to zero). The adopted criteria were $0.25f_{x1}$ and $0.25f_{y1}$ for model BH-1, $0.175f_{x2}$ and $0.175f_{y2}$ for model BH-2, and $0.12f_{x3}$ and $0.12f_{x3}$ for model BH-3.

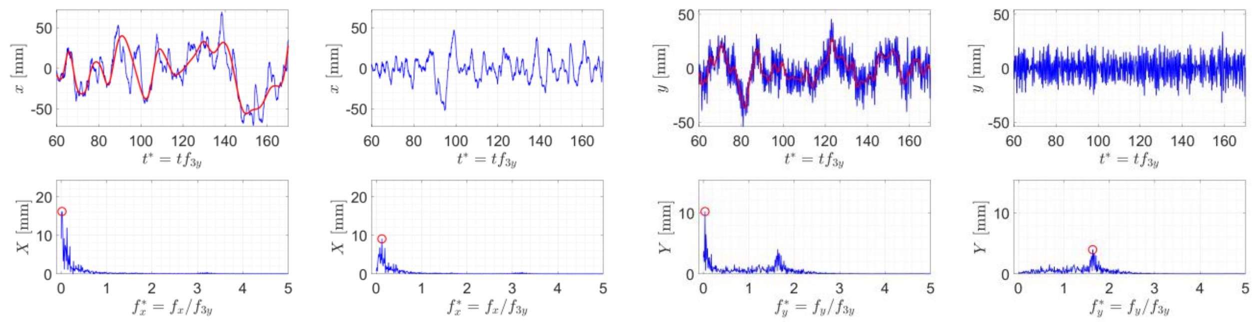
The rationale behind discounting the low-frequency content is that this energy component is not regarded as part of the VIV dynamics. Instead, it is perceived as an outcome of long-periodic oscillations exhibited by the pipe around its mean equilibrium configuration, which are a consequence of slow drag fluctuations. As the pipe is flexible, the drag induces a streamwise angle of inclination along its length. This, in turn, reduces the projection of the pipe's area facing the stream, thereby reducing drag and causing a reduction in the streamwise angle of inclination. Consequently, the projected area of the pipe to the incoming stream is increased, thus increasing drag and the angle of inclination once more. This results in the generation of oscillations of very large periods. It is crucial to highlight that this phenomenon does not align with the usual doubled streamwise frequency resulting from vortex shedding, but rather with the bending flexibility of the pipe. This step is illustrated in the subplots on the second line of Figure 4(a) and Figure 4(b), where the selected discretized steady-state signal subsequent to the filtering procedure, accompanied by its respective amplitude spectrum, are depicted on the left and right sides, respectively.

To elucidate the discounting of the low-frequency content further, Figure 5(a) and Figure 5(b) illustrate the effect of filtering out the low-frequency component from the selected portion of the time series. In each chart, the selected portion of the time series and the respective amplitude spectra are represented in the first and second subplots, respectively, on the left side. The mean value and the residual trend have been discounted, yet the low-frequency content remains embedded, as illustrated by the red lines. Conversely, the same portion of the time series and the corresponding amplitude spectra, displayed on the right side of each chart, demonstrate the exclusion of low-frequency content. As can be observed, the distinction is apparent, as the amplitude spectra on the left side contain dominant peaks of energy at very low frequencies (marginal to zero), whereas the spectra related to the portion of the signal with the low-frequency content filtered out demonstrate that the frequencies of oscillation associated with VIV become the primary peak of response. The statistics along the length of the models were constructed using the discrete signals resulting from the aforementioned steps.



(a) Streamwise direction. (b) Crosswise direction.

Figure 4: Time series from the tip of model BH-3 in pure towing test at $U = 0.52$ m/s ($U^*_{3y} = 13.71$). Complete time series (subplot above), selected range of time series after filtering (subplot below on the left) and respective amplitude spectrum (subplot below on the right).



(a) Streamwise direction. (b) Crosswise direction.

Figure 5: Time series from the tip of model BH-3 in pure towing test at $U = 0.52$ m/s ($U^*_{3y} = 13.71$). Selected piece of the signal (subplots above), respective amplitude spectra (subplots below); low frequency content embedded (on the left); low frequency content filtered out (on the right).

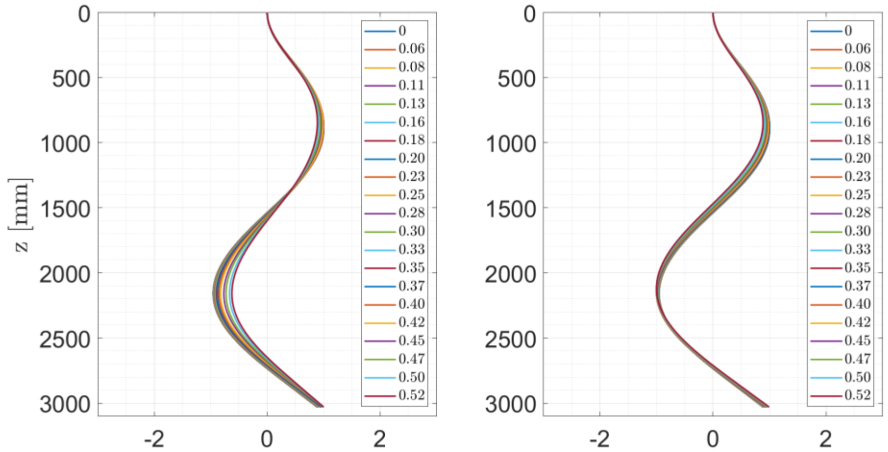
The second part of the data analysis procedure implements Galerkin's projection on spatial functions, ψ_n, on the discretized steady state coordinate time series with null average over time. A projection set was created for each experimental model and every velocity tested, using the in-house mathematical model based on the experimental models' as-built parameters. Each set contains sixteen complex-conjugated pairs of spatial functions, from where eight were selected for each direction. Modal amplitudes A_n referring to the n^{th} natural mode of vibration, were obtained at instants of time t_j, as presented in Eqs. (1) and (2), for both the streamwise and crosswise directions, respectively. The in-line and crosswise coordinates displacements along the pipes' lengths are $x(s,t_j)$ and $y(s,t_j)$, respectively, and $\psi_n(s)$ is the spatial projection function of the n^{th} natural mode of vibration, whereas s refers to arc length positions. The same procedure can be seen in Franzini *et al*. (2015, [16]) and Salles and Pesce (2019, [36]).

$$A_{n\,x}(t_j) = \frac{\int_0^L x(s,t_j)\,\psi_n(s)\,ds}{\int_0^L (\psi_n(s))^2\,ds} \tag{1}$$

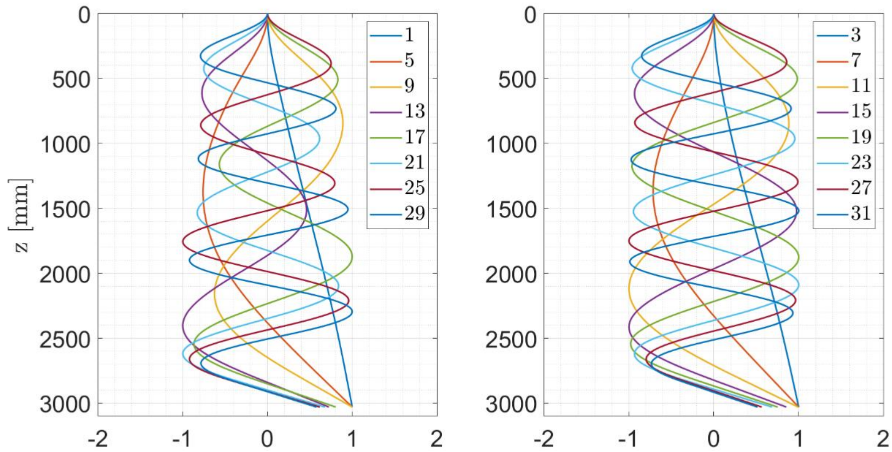
$$A_{n\,y}(t_j) = \frac{\int_0^L y(s,t_j)\,\psi_n(s)\,ds}{\int_0^L (\psi_n(s))^2\,ds} \tag{2}$$

Figure 6(a) depicts the set of projection functions for the third natural mode of vibration, as a function of the external flow velocity (left: streamwise; right: crosswise). As can be observed, the streamwise function is the most significantly affected by the drag effect. The second subplot, Figure 6(b), depicts the sixteen projection functions (eight for each direction) used in the analysis at the maximum velocity tested. The legends indicate the chosen function (absent numbers are the complex-conjugate counterparts). Figure 7 depicts the obtained modal amplitude time series accompanied by their corresponding amplitude spectra, for the BH-3 model at $U = 0.52$ m/s ($U^*_{3y} = 13.71$).

Subsequently, the modal amplitude responses, normalized with respect to the external diameter, are presented as a function of the modal reduced velocity in two distinct yet complementary approaches. In the first approach, the respective amplitude spectra are presented in a color scheme, which is considered a more comprehensive representation of the dynamics, as it depicts the amplitude components across the entire frequency band. The second approach, which is more traditional and commonly used in VIV analysis, provides qualitative inspections in a straightforward manner, calculating the equivalent amplitude by its root mean square (RMS) times $\sqrt{2}$, and then referencing it to the frequency of the dominant peak.

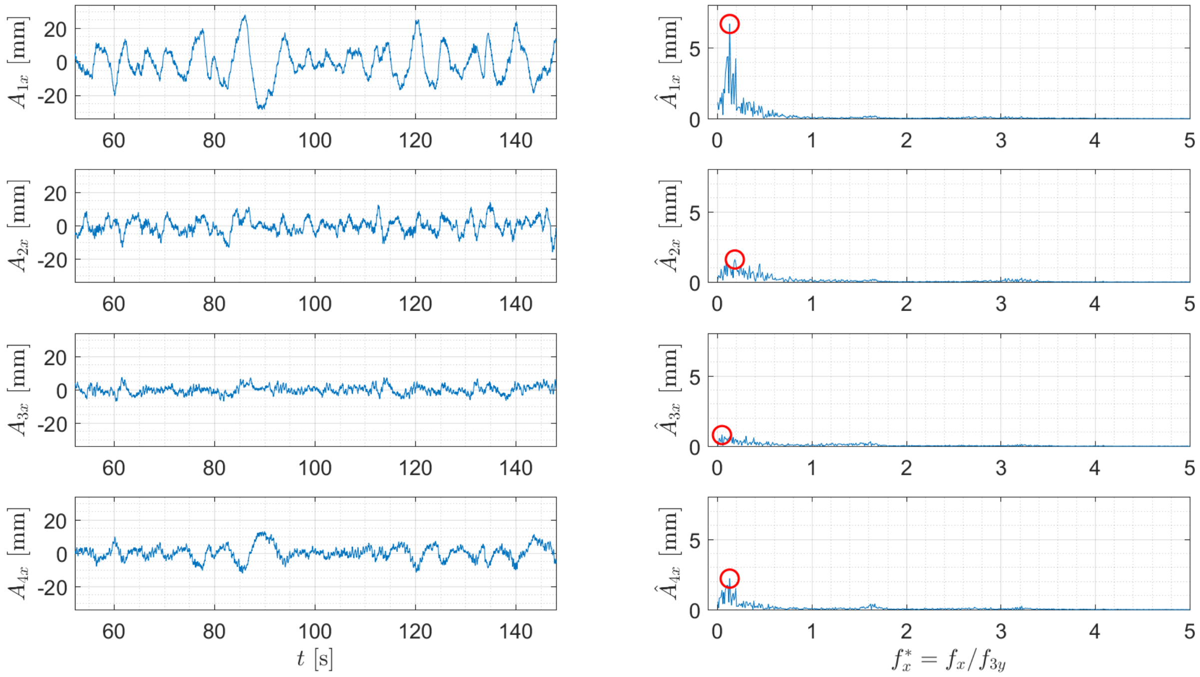


(a) Influence of the external flow velocity on the projection function for the 3^{rd} natural mode of vibration. Left: streamwise direction; right: crosswise direction. Legends indicate the flow velocity in m/s.

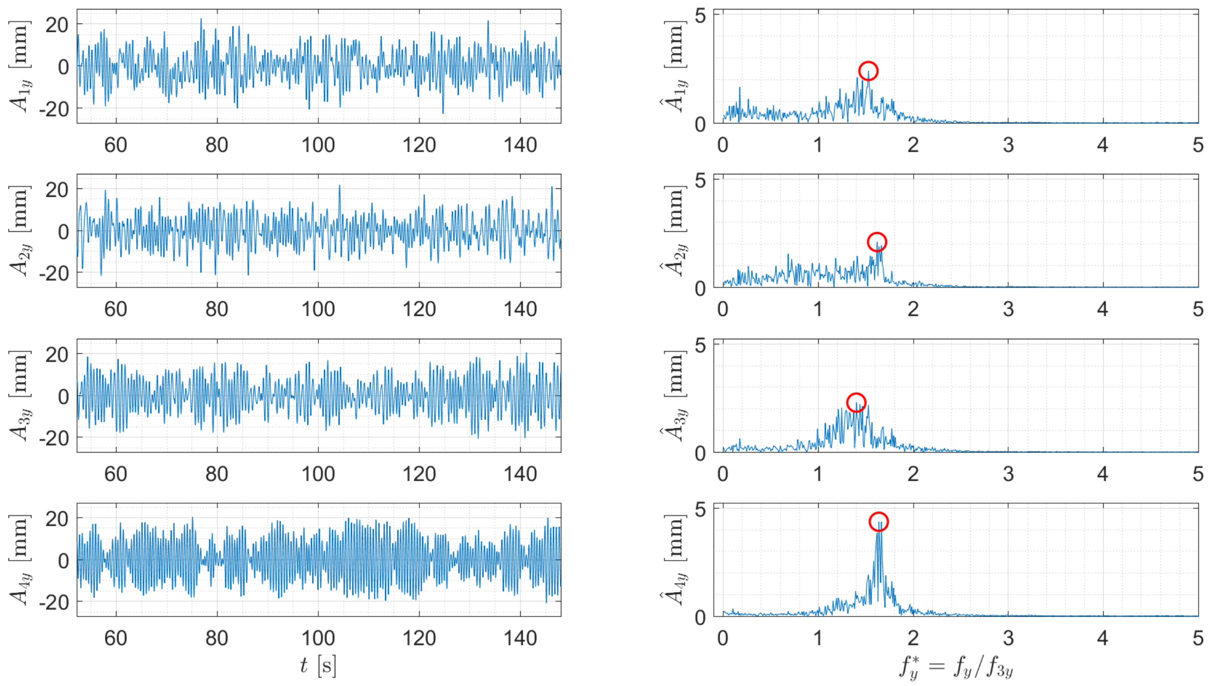


(b) Set of projection functions used for the maximum external flow velocity, $U = 0.52$ m/s.

Figure 6: Streamwise (left) and crosswise (right) spatial functions. Experimental model BH-3.



(a) Streamwise direction



(b) Crosswise direction.

Figure 7: Modal amplitude time series and the respective amplitude spectra. Red circles indicate the dominant amplitude peak. Towing test with model BH-3 at $U = 0.52$ m/s ($U_{3y}^* = 13.71$).

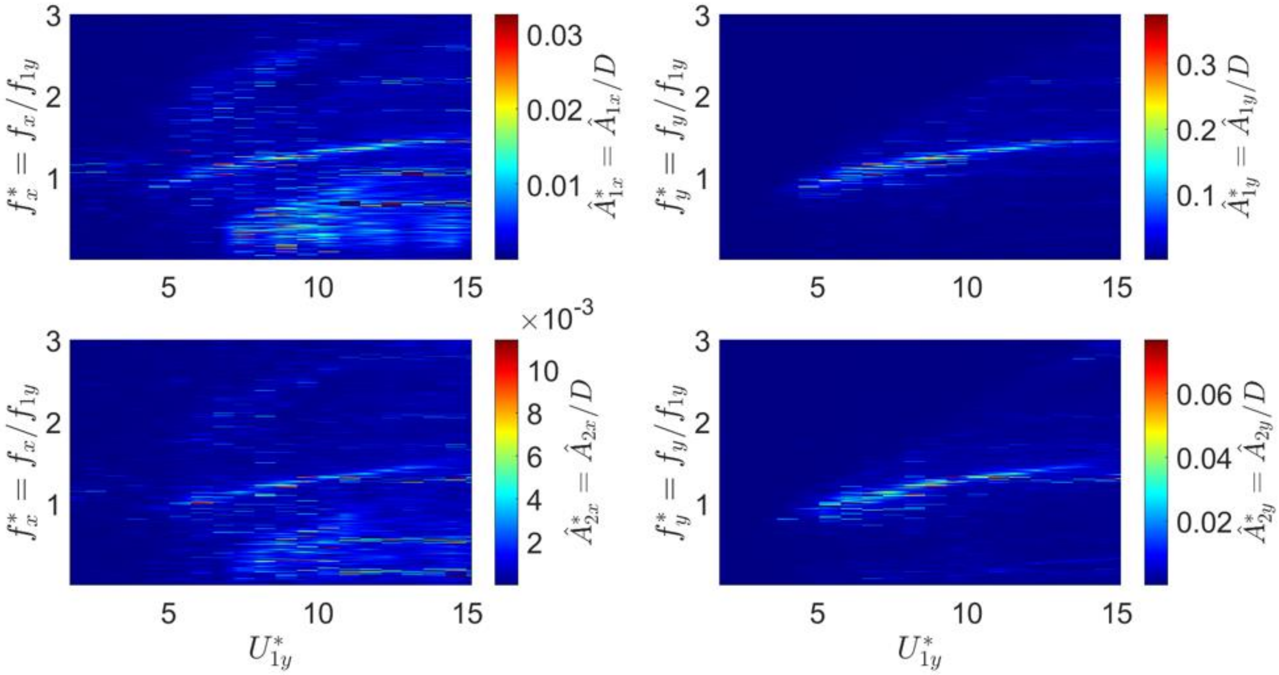
3. Results

Further on, a selection of the most interesting findings is discussed, with the reader directed to [11] for the comprehensive set of responses pertaining to this investigation. In the next modal responses, the left side presents the streamwise dynamics, while the right side shows the crosswise one. The modal amplitudes are normalized by the external diameter of the pipe and the modal frequencies are normalized by the respective frequency of interest of each experimental model.

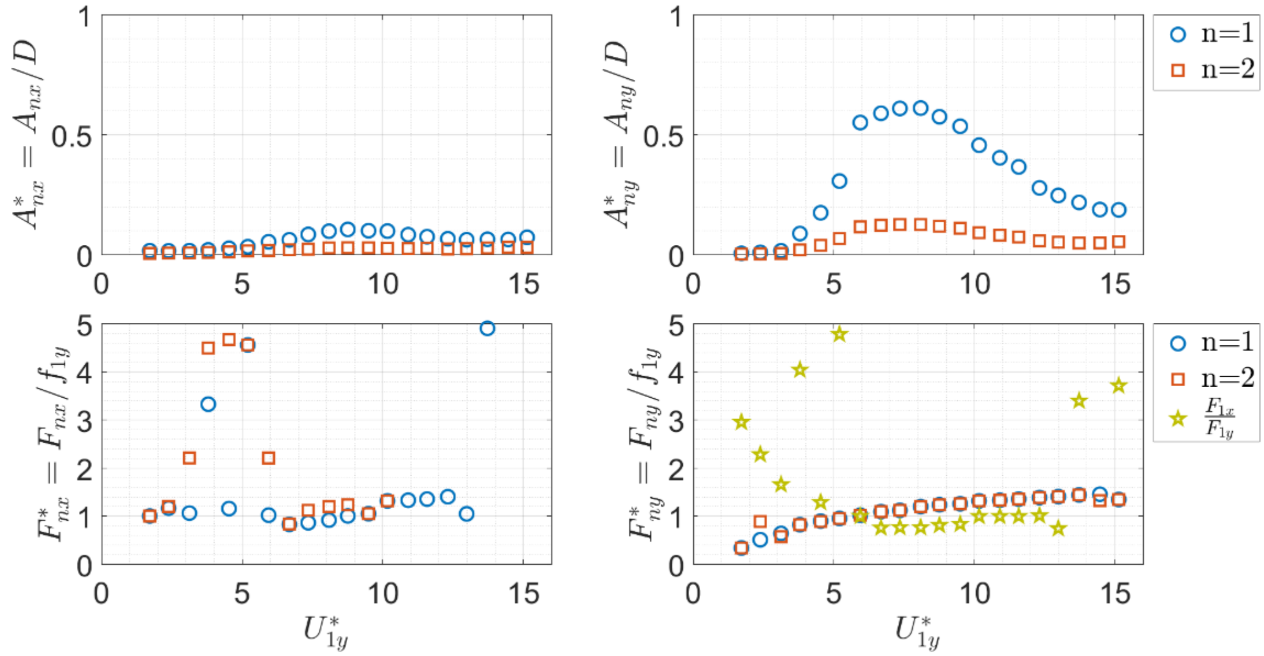
3.1. Pure VIV

The modal amplitude spectra obtained from model BH-1 (Figure 8a) demonstrate that the primary amplitude contribution is attributable to the first crosswise natural mode of vibration, which exhibits a trace of amplitudes within a narrow frequency band throughout the entire velocity test range. This observation aligns with the anticipated dynamics of the model, which was designed to exhibit such behavior by having the value of its first crosswise natural frequency settled to match the vortex-shedding frequency in the usual VIV lock-in region ($5.0 \leq U_{1y}^{*} \leq 8.0$). The second crosswise mode also exhibits a comparable trace of response, albeit with lower amplitudes, within the same constrained frequency band. The similarity in frequency response with varying amplitudes suggests that one mode is dominant over the other. In the present case, the first crosswise mode is dominant over the second crosswise mode. In the streamwise direction, very weak traces of amplitude are present at $U_{1y}^{*} = 2.5$, indicating the lock-in of the first mode. In further reduced velocities, the first and second modes exhibit traces of amplitude that are similar to the crosswise response, although with smaller magnitudes. Components of amplitude dispersed in a wide frequency band are present throughout the entire range of tests. Such amplitude dispersion is to be expected in the streamwise direction due to the effects of complex streamwise vortex-shedding fluctuations and long periodic oscillations caused by drag fluctuations.

The analysis of the modal responses as monochromatic signals (Figure 8b) facilitates the visualization of the first crosswise mode initial branch of response, followed by an amplitude jump to the respective upper branch of response. In contrast with the VIV responses of rigid cylinders and pipes, there is no discernible second amplitude jump (from the upper to the lower branch). Instead, there is an extension of the resonance to higher reduced velocities and a continuous amplitude decrease until the desynchronization regime is attained. This distinctive behavior is attributed to the bending flexibility of the model, which may facilitate accommodation of the structure to the varying vortex-shedding pattern. All the other natural modes, in both the crosswise and streamwise directions, exhibit amplitudes in the same reduced velocity region as the first crosswise natural mode upper branch. Despite not being in their own resonance region, they are still excited by a mechanism that transfers energy. Additionally, their respective frequencies follow the frequency response of the first crosswise mode, thereby reinforcing the conclusion that this mode is dominant. This can be better visualized in Figure 9, which illustrates the dynamics along the model's length at the peak of the upper branch, approximately at $U_{1y}^{*} \approx 8.0$, through its trajectory projections, the respective standard deviation, and the respective amplitude spectra, in both the directions. In the crosswise direction, the amplitude distribution exhibits a clear and defined trace of amplitudes at approximately $f_{1y}^{*} \approx 1.2$, where amplitudes of the first and second natural modes coexist. Conversely, the streamwise direction presents a scattered amplitude distribution in basically three frequencies, at the same as the crosswise direction, weak traces at the double value of it, and at a range of low values up to $f_{1y}^{*} \approx 0.7$, mostly composed of second streamwise mode amplitudes. Modal orbits onto the XY plane, at $U_{1y}^{*} \approx 8.0$, are depicted by Figure 10. The orbit projection forms an ellipsoidal shape, which is characteristic of a 1:1 frequency ratio, i.e., $F_{1x}/F_{1y} = 1.0$. The dual resonance regime, identified by the frequency ratio $F_{1x}/F_{1y} \approx 2.0$, is represented by the star shaped markers at Figure 8(b). This specific frequency ratio appears at $U_{1y}^{*} \approx 2.5$, which is the theoretical reduced velocity for the resonance of the first streamwise natural mode.



(a) Modal amplitude spectra as function of the dimensionless frequencies, f_x^* and f_y^*, and of the reduced velocity, U_{1y}^*.



(b) Modal amplitudes and the respective dominant frequencies as function of the reduced velocity, U_{1y}^*.

Figure 8: Experimental model BH-1 modal responses. In both charts, columns on the left and right are for the streamwise and crosswise directions, respectively.

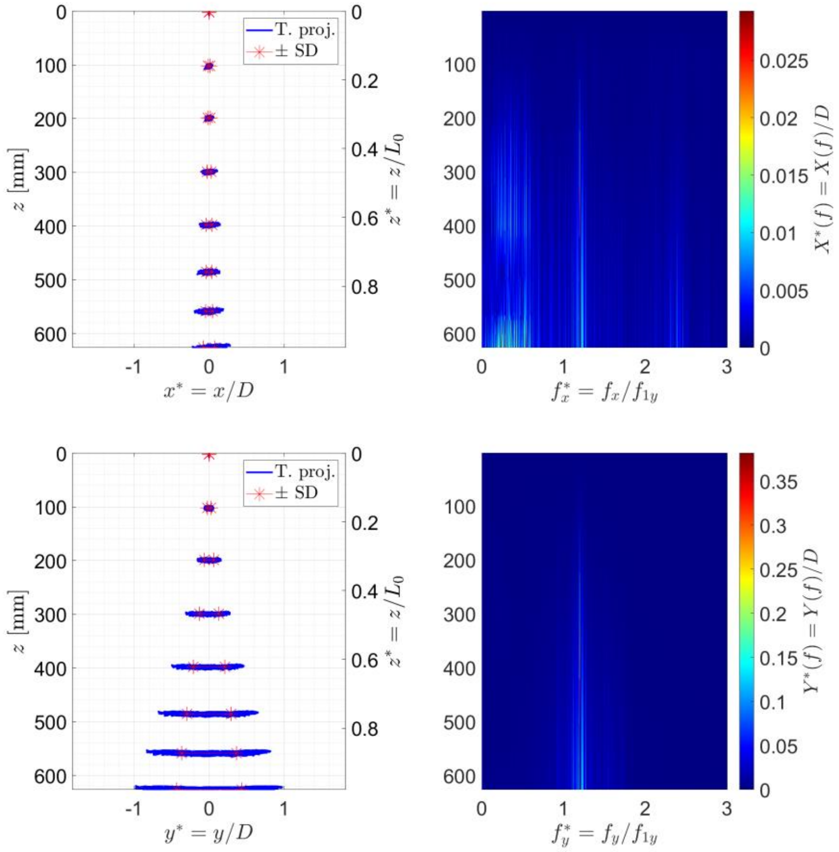


Figure 9: (Left) Trajectory projections (in blue) and standard deviation (in red). (Right) respective amplitude spectra of all targets along the model's length. Pure towing test at $U = 0.28$ m/s ($U_{1y}^* = 8.09$).

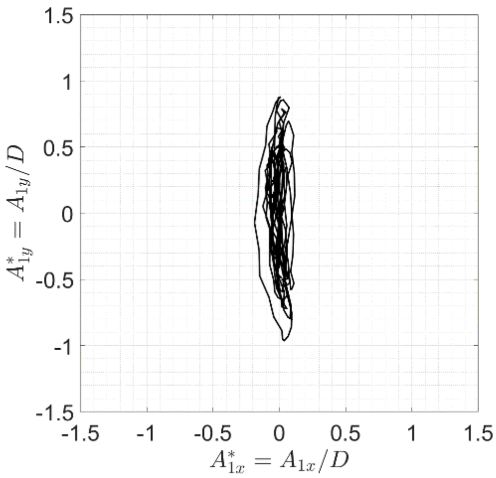
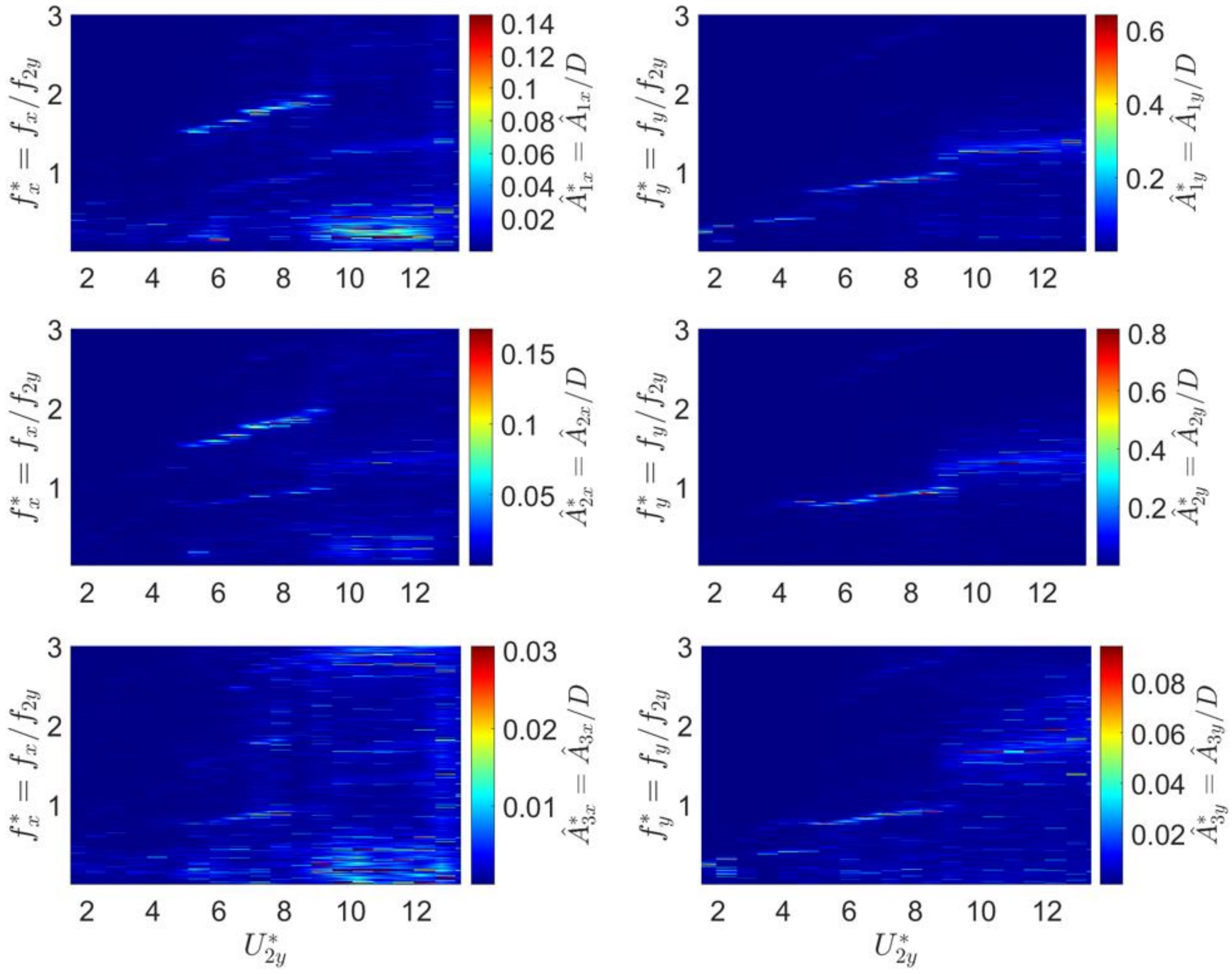


Figure 10: Modal orbits on the XY plane. Pure towing test at $U = 0.28$ m/s ($U_{1y}^* = 8.09$).

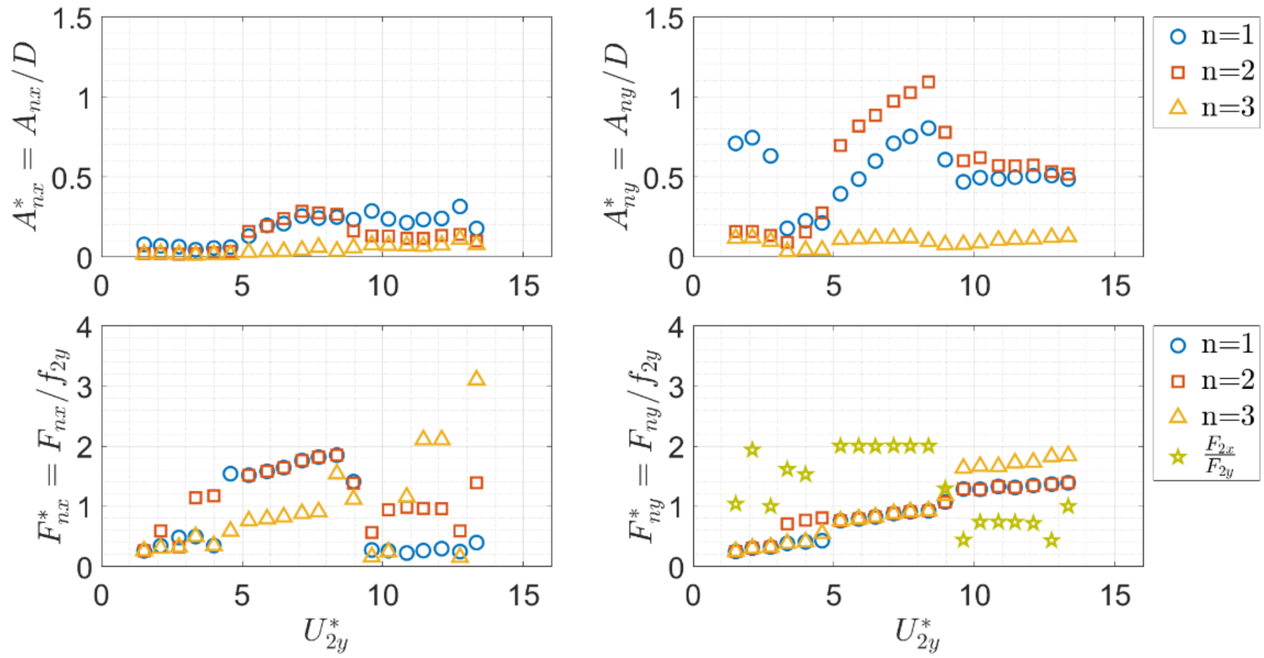
The response of the BH-2 model is more complex. In Figure 11(a), the modal amplitudes spectra response of its dynamics depict resonances of the first and second crosswise modes delineated by clear traces of amplitudes in constrained frequency bands. Such narrow frequency arrangement is kept until $U_{2y}^* \approx 8.5$, from where the dynamics becomes characterized by amplitude components originating from a broader frequency range. The amplitudes of the third crosswise mode and of the three streamwise modes remain at a relatively low level throughout the entire test range within a broader frequency range. Moreover, the amplitudes of the first and second streamwise modes exhibit notable similarities with regard to their magnitudes, frequency bands with broader spectrum, and reduced velocity regions of appearance. Additionally, a relatively strong amplitude trace with low frequencies is discernible in the first streamwise mode response at the final velocities tested.

This feature may be related to long periodic drag fluctuations around the mean equilibrium position. Despite the majority of this effect having been removed from the data, as previously detailed in Section 2.2, a residual portion remains embedded in the signals. Complementarily, the third streamwise mode does not present any well-defined trace of amplitude but rather small components in a very sparse spectrum.

The present dynamics demonstrates not only the mechanism of energy transfer from the dominant crosswise mode to others, but also a transition of modal resonances as the flow velocity is increased. Figure 11(b) illustrates that. At the initial velocities tested, $1.0 \leq U_{2y}^{*} \leq 3.0$, the model exhibits the upper branch of response related to the first crosswise mode of vibration with amplitudes reaching $0.7D$. This results in a weak excitation of the second and third crosswise modes, thus in this region the dominance is attributed to the first mode. As the flow velocity is increased, the resonance of the first crosswise mode abruptly ceases to exist, being succeeded by the resonance of the second crosswise mode, which becomes the dominant one. This transition occurs at the region $3.0 \leq U_{2y}^{*} \leq 5.0$, where the initial branch of the second crosswise mode is observed. Subsequently, an amplitude jump occurs, resulting in the emergence of the second mode's upper branch of response, which is maintained from $U_{2y}^{*} = 5.0$ until approximately $U_{2y}^{*} = 8.5$. In this region, amplitudes varying from 0.7 to $1.1D$. Figure 12 illustrates the model's dynamics along its length at $U_{2y}^{*} = 8.37$. The crosswise trajectory projections evidence the amplitude distribution in a clear second mode shape and the respective amplitude spectra depict them within a strict frequency band at almost exactly $f_{2y}^{*} = 1.0$. As previously outlined in Section 2.1, it was the objective to observe the second crosswise natural mode exhibiting the primary resonance in the present experimental model. Back to Figure 11(b), a second jump in amplitude occurs at $8.5 \leq U_{2y}^{*} \leq 10.0$, which is contrary to the previous model's behavior and aligns with the dynamics of rigid cylinders. This indicates the onset of the lower branch of response of the second crosswise mode, which persists up to the final velocities tested. It is noteworthy that the first crosswise mode is significantly excited by the second mode's resonance. The former reaches amplitudes of approximately $0.8D$ in the latter's upper branch and maintains similar amplitudes in its lower branch, a remarkable qualitative resemblance. In the streamwise direction, the first and second modes are also stimulated in the region of the second crosswise mode upper branch, albeit at a frequency that is twice as high. In fact, throughout the entire primary resonance region, $5.0 \leq U_{2y}^{*} \leq 8.5$, a dual resonance regime is attained between the second streamwise and crosswise modes, as illustrated by the curve with star markers, which indicates $F_{2x}/F_{2y} \approx 2.0$. Moreover, the projections of modal orbits onto the XY plane in Figure 13 demonstrate that, despite with varying amplitudes, the dual resonance phenomenon is attained between all natural modes, not merely the dominant and its counter-orthogonal part, as evidenced by the Lissajous-like patterns.



(a) Modal amplitude spectra as function of the dimensionless frequencies, f_x^* and f_y^*, and of the reduced velocity, U_{2y}^*.



(b) Modal amplitudes and the respective dominant frequencies as function of the reduced velocity, U_{2y}^*.

Figure 11: Experimental model BH-2 modal responses. In both charts, columns on the left and right are for the streamwise and crosswise directions, respectively.

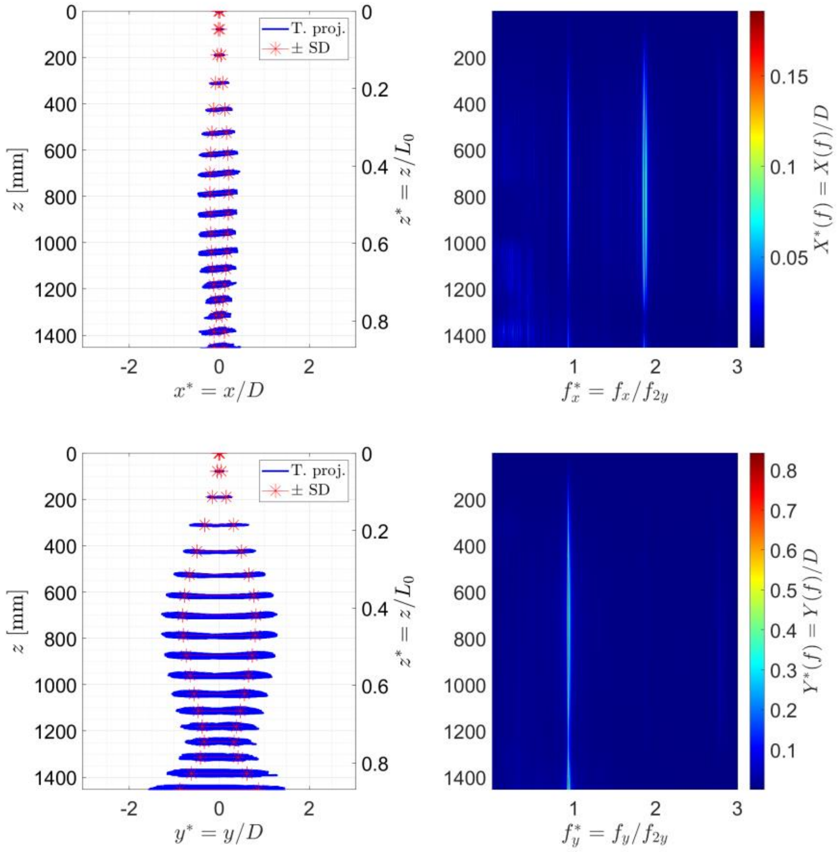


Figure 12: (Left) Trajectory projections (in blue) and standard deviation (in red) and (right) respective amplitude spectra of all targets along the model's length. Towing test at $U = 0.33$ m/s ($U_{2y}^* = 8.37$).

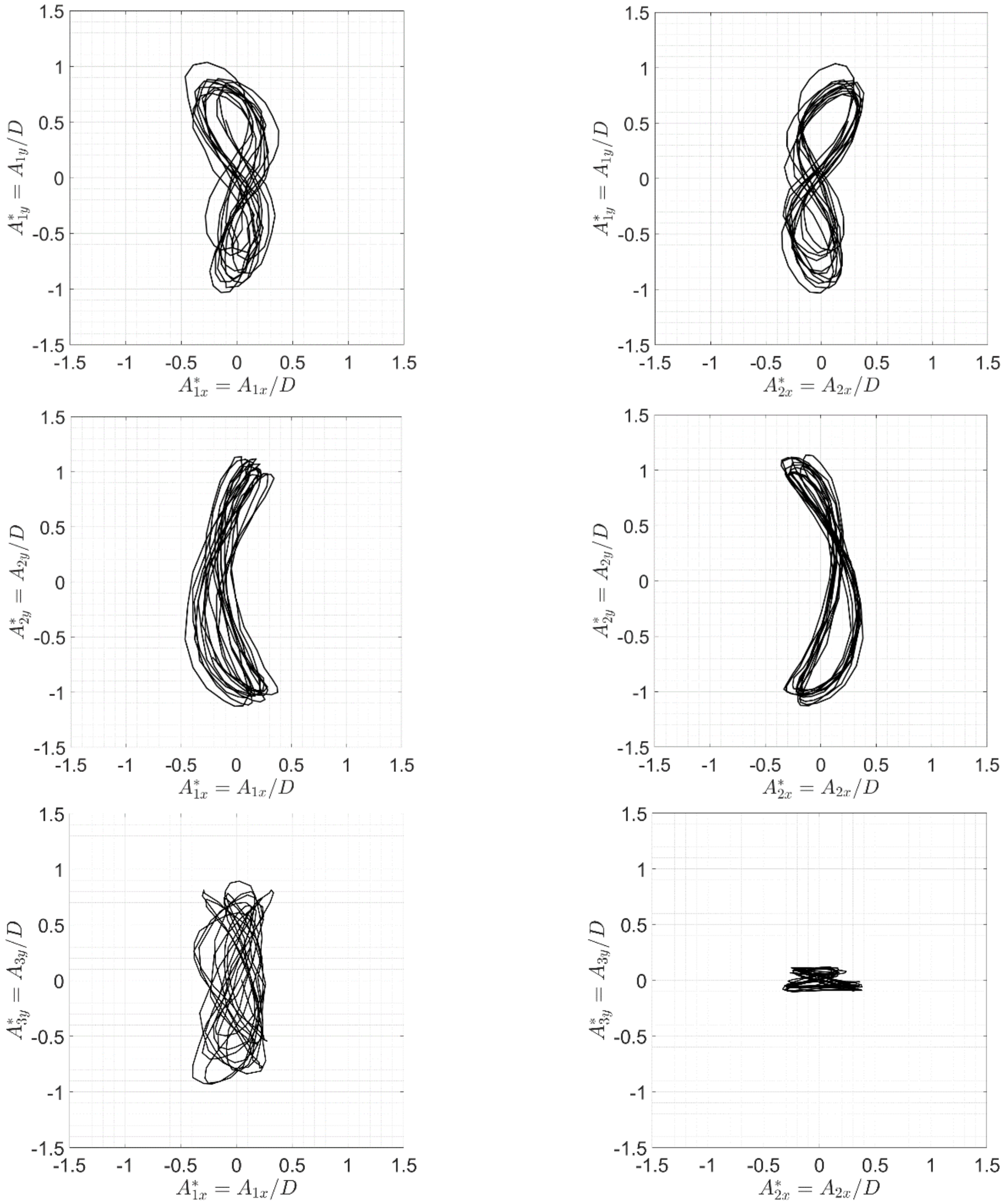
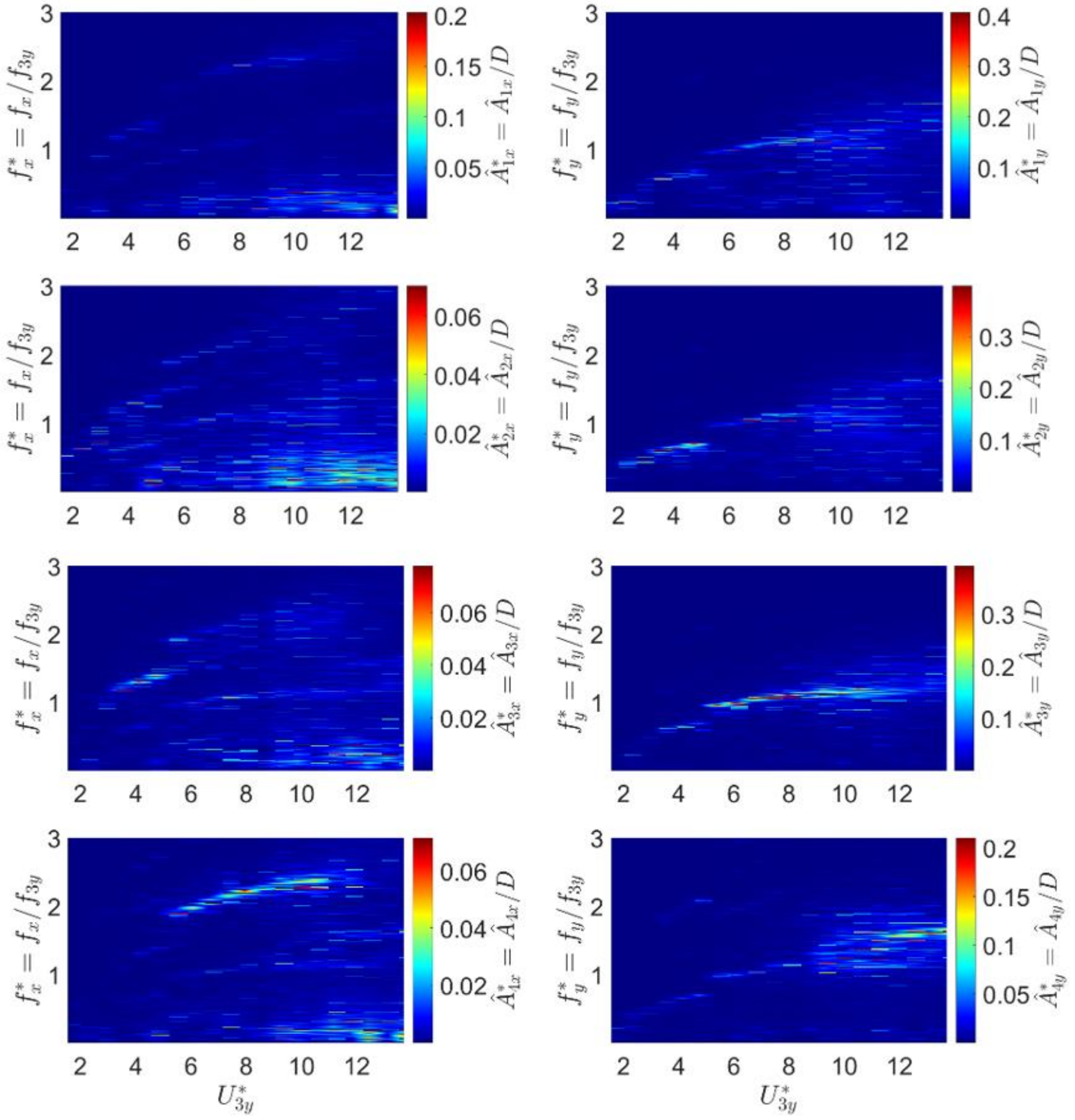


Figure 13: Modal orbits on the XY plane. Pure towing test at $U = 0.33$ m/s, $U^*_{2y} = 8.37$. (On the left) A^*_{1y}, A^*_{2y} and A^*_{3y} vs. A^*_{1x}. (On the right) A^*_{1y}, A^*_{2y} and A^*_{3y} vs. A^*_{2x}.

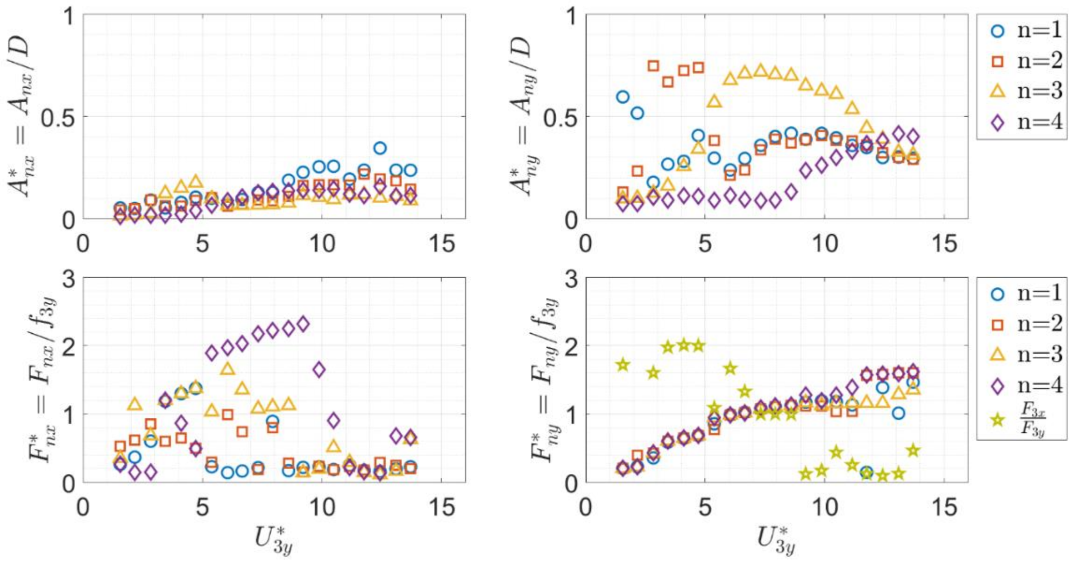
A similar dynamic behavior is revealed by the experimental model BH-3, which demonstrates the mechanism of energy transfer from a dominant mode to others, as well as transitions of upper branches of response. As illustrated in Figure 14(a), the modal amplitude spectra indicate that the third natural mode exhibits the primary traces of resonance in the crosswise direction. The amplitudes are constrained within a narrow frequency band and sustained over the reduced velocity region $5.0 \leq U^*_{3y} \leq 10.0$. Similarly, as with the preceding models, this one was designed to exhibit this feature, with the third natural mode considered the mode of interest. The remaining crosswise natural modes reflect this primary trace in their responses, indicating that they are being stimulated by the second crosswise mode. As the test velocity is increased beyond approximately $U^*_{3y} \approx 9.0$, all crosswise modes present amplitudes composed by dispersed frequency contents. Furthermore, the energy transfer mechanism occurs for the streamwise modes as well, as evidenced by the blurry yet discernible amplitude traces depicted at the reduced velocity region of primary resonance of the second crosswise mode. As expected, the majority of the amplitude exhibited by the streamwise modes originates from components distributed across a broad frequency spectrum throughout the entire test range. Interestingly, the fourth streamwise mode displays a relatively strong trace of amplitude at $5.0 \leq U_{3y}^{*} \leq 10.0$, encompassing frequencies that are twice as high as those of the dominant crosswise mode. This feature was also observed in the previous model.

The transition between response branches is clearly visible in Figure 14(b). At the initial velocities tested, the crosswise response exhibits the first mode as the dominant one, which presents its upper branch of response coexisting with the second mode's initial branch up to $U_{3y}^{*} = 2.0$. Subsequently, as the velocity is increased, the first mode's upper branch abruptly ceases to exist as the second mode becomes the dominant one, which is marked by an amplitude jump from its initial to the upper branch of response, occurring at $2.5 \leq U_{3y}^{*} \leq 5.0$. The first mode persists in presenting amplitudes within this region after its own resonance, which is understood to be a transfer energy mechanism. Furthermore, this reduced velocity region also exhibits the emergence of the third mode's resonance. Figure 15 illustrates the dynamic behavior along the model's length at $U_{3y}^{*} = 4.1$. In the crosswise direction, an amplitude distribution consistent with the second natural mode is exhibited in a narrow frequency band centered at $f_{3y}^{*} \approx 0.7$. In the streamwise direction, the amplitude spectra demonstrate a distinct and concentrated third mode shape amplitude distribution at $f_{3y}^{*} \approx 1.4$, accompanied by blurry and weak traces of amplitude within lower frequency bands. Referring back to Figure 14(b), at $U_{3y}^{*} = 5.0$, a second transition of resonance occurs. The amplitudes of the third mode undergo a sudden transition, jumping from their initial branch of response to the upper branch. Conversely, the amplitudes of the second mode's upper branch abruptly cease. The third mode assumes dominance in the dynamics, with its upper branch remaining at $5.0 \leq U_{3y}^{*} \leq 10.0$. It is noteworthy that the amplitudes of the first and second modes are remarkably similar in this velocity region, as they are stimulated by the dominant mode (the third). It is pertinent to highlight that the first mode is subjected to a second excitation, which occurs well beyond its respective upper branch region. Figure 16 illustrates the dynamics along the model's length at the reduced velocity $U_{3y}^{*} = 7.32$. In the crosswise direction, a distinct third natural mode of vibration is depicted by both the trajectory projections and the amplitude spectra, which is very constricted at $f_{3y}^{*} \approx 1.05$ and clearly exhibits two nodal points. The spectra in the streamwise direction illustrate the presence of a pronounced and discernible fourth mode shape at $f_{3y}^{*} \approx 2.1$, coexisting with components of lower modes constrained to lower frequency bands. At the final velocities tested, Figure 14(b) illustrates that the third mode's upper branch displays a gradual decline in amplitude, devoid of any apparent discontinuity, until a desynchronization regime is reached. This feature is notably repeated by the first and second modes. Furthermore, the onset of the fourth natural mode is clearly discernible at $U_{3y}^{*} = 8.0$ onwards. Practically, over the entire range of tests the crosswise modal frequencies are maintained constrained, irrespective of the natural mode that is the dominant. In the streamwise direction, no discernible prominence of any natural mode is observed, with their amplitudes exhibiting similar and relatively low magnitudes across the entire range of tests.

It is noteworthy that the dual resonance regime between the third streamwise and crosswise natural modes is identified in the region $2.5 \leq U_{3y}^{*} \leq 5.0$. This coincides with the second crosswise mode's upper branch, as illustrated by the star markers in Figure 14(b). This is corroborated by Figure 17, which depicts the projection of modal orbits at $U_{3y}^{*} = 4.10$, encompassing a combination of every natural mode. The third streamwise natural mode exhibits eight-shaped orbits with every other crosswise mode, providing a visual representation of dual resonance. Figure 18 presents a similar panel with the projection of modal orbits at $U_{3y}^{*} = 7.32$, specifically for the fourth streamwise natural mode. This mode exhibits dual resonance regimes with all the crosswise modes at $5.0 \leq U_{3y}^{*} \leq 10.0$, as evidenced by the eight-shape orbits and the doubled oscillation frequency value in comparison to the other crosswise modes. This evidence is noteworthy. The flexibility of the system allows not only the transfer of energy from a dominant mode to other modes but also, as a consequence of the narrowing of modal frequencies within a frequency band, the occurrence of multiple simultaneous dual resonance phenomena.



(a) Modal amplitude spectra as function of the dimensionless frequencies, f_x^* and f_y^*, and of the reduced velocity, U_{3y}^*.



b) Modal amplitudes and the respective dominant frequencies as function of the reduced velocity, U_{3y}^*.

Figure 14: Experimental model BH-3 modal responses. Columns on the left and right are for the streamwise and crosswise directions, respectively.

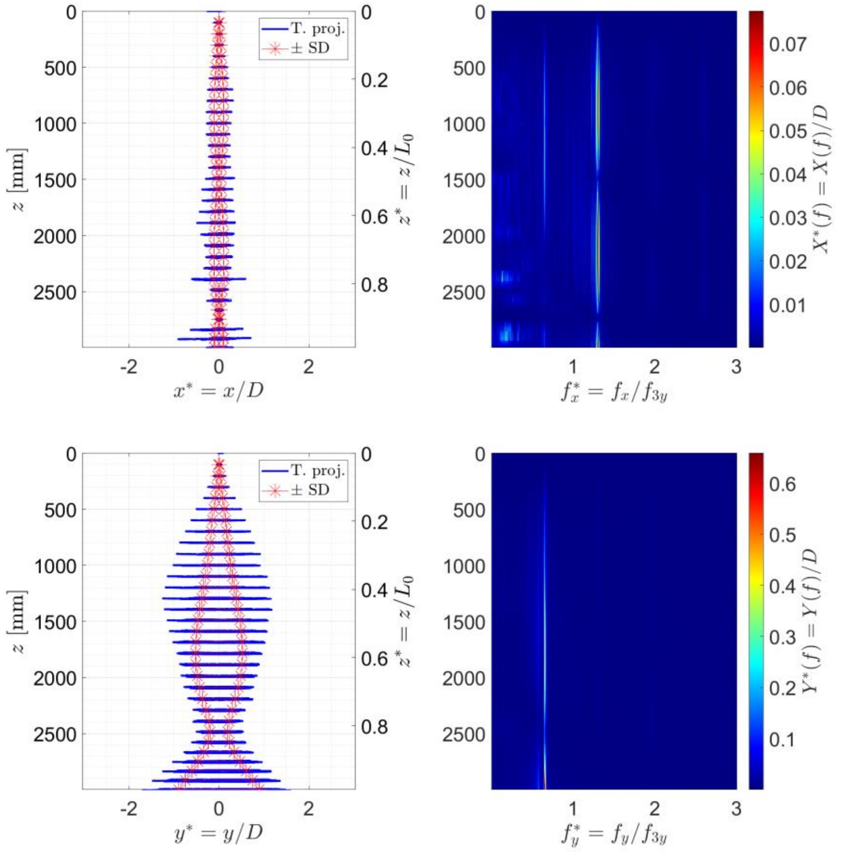


Figure 15: (Left) Trajectory projections (in blue) and standard deviation (in red) and (right) respective amplitude spectra of all targets along the model's length. Towing test at $U = 0.16$ m/s ($U_{3y}^* = 4.10$).

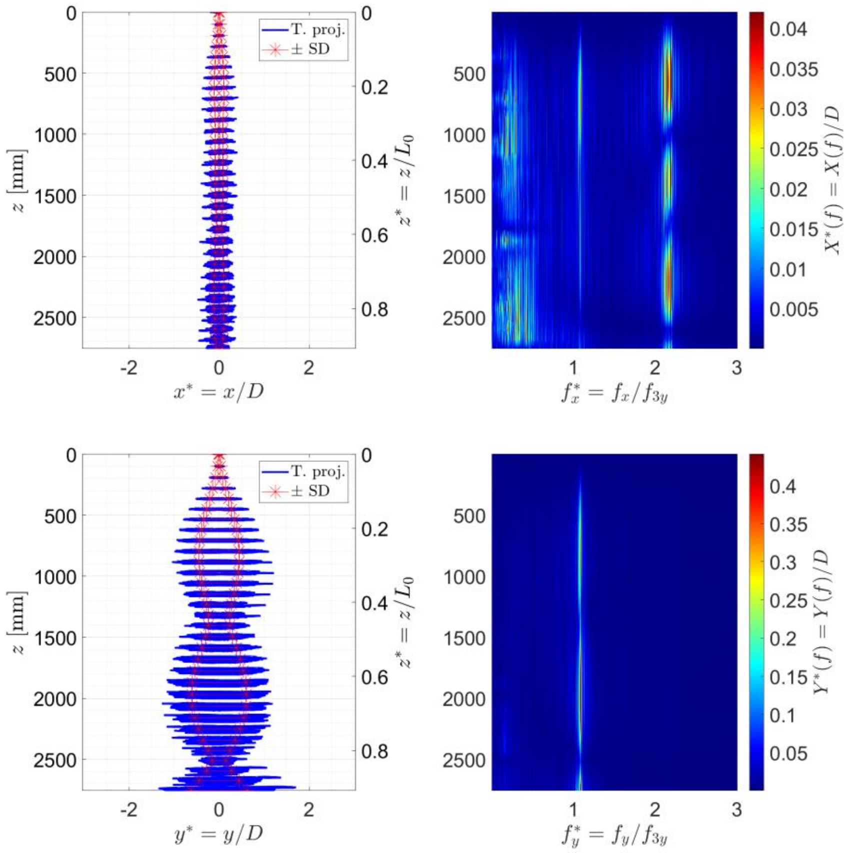


Figure 16: (Left) Trajectory projections (in blue) and standard deviation (in red) and (right) respective amplitude spectra of all targets along the model's length. Towing test at $U = 0.28$ m/s ($U_{3y}^* = 7.32$).

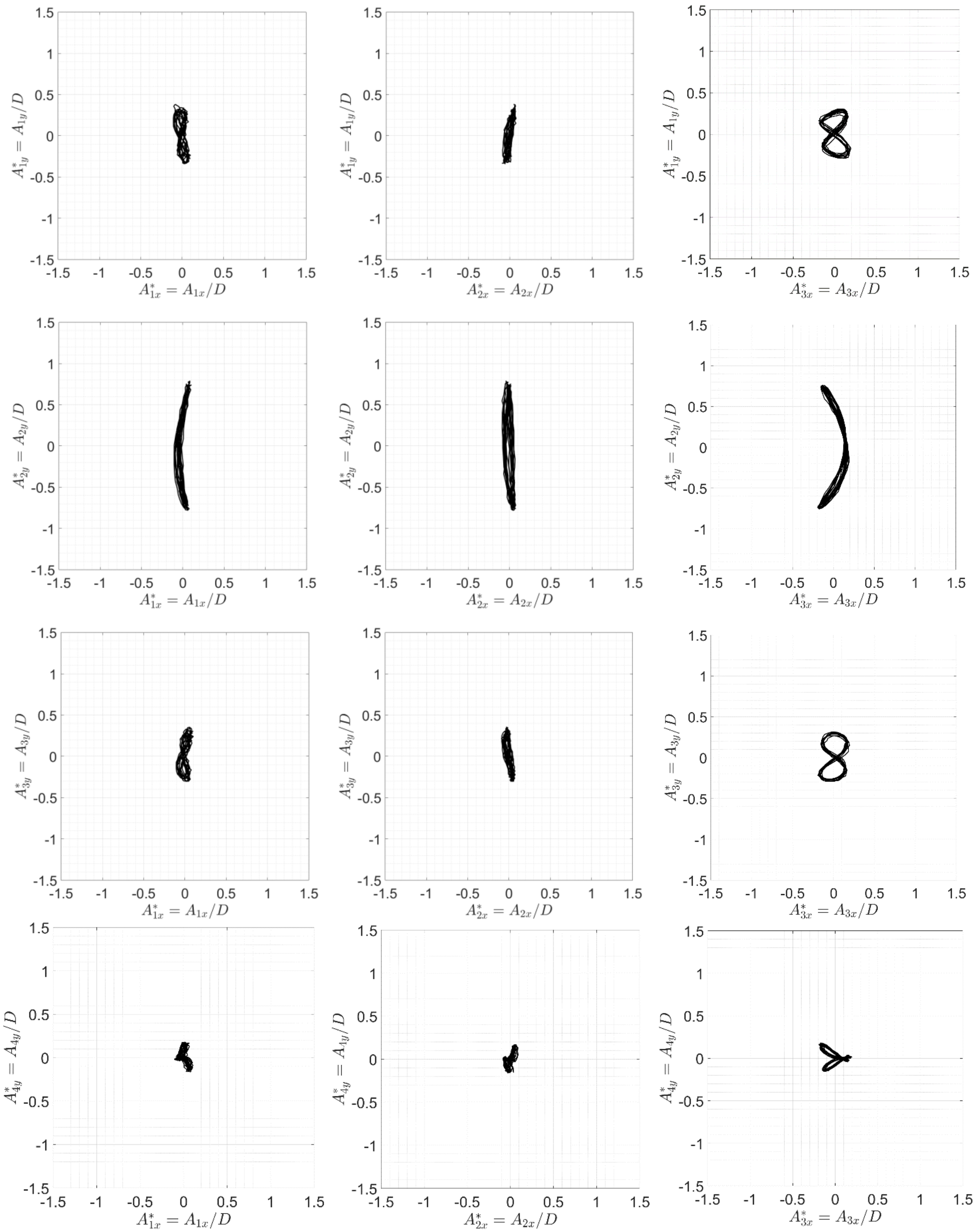


Figure 17: Modal orbits on the XY plane. Pure towing test at $U = 0.16$ m/s ($U^*_{3y} = 4.10$). (On the left) A^*_{1y}, A^*_{2y}, A^*_{3y} and A^*_{4y} vs. A^*_{1x}. (In the middle) A^*_{1y}, A^*_{2y}, A^*_{3y} and A^*_{4y} vs. A^*_{2x}. (On the right) A^*_{1y}, A^*_{2y}, A^*_{3y} and A^*_{4y} vs. A^*_{3x}.

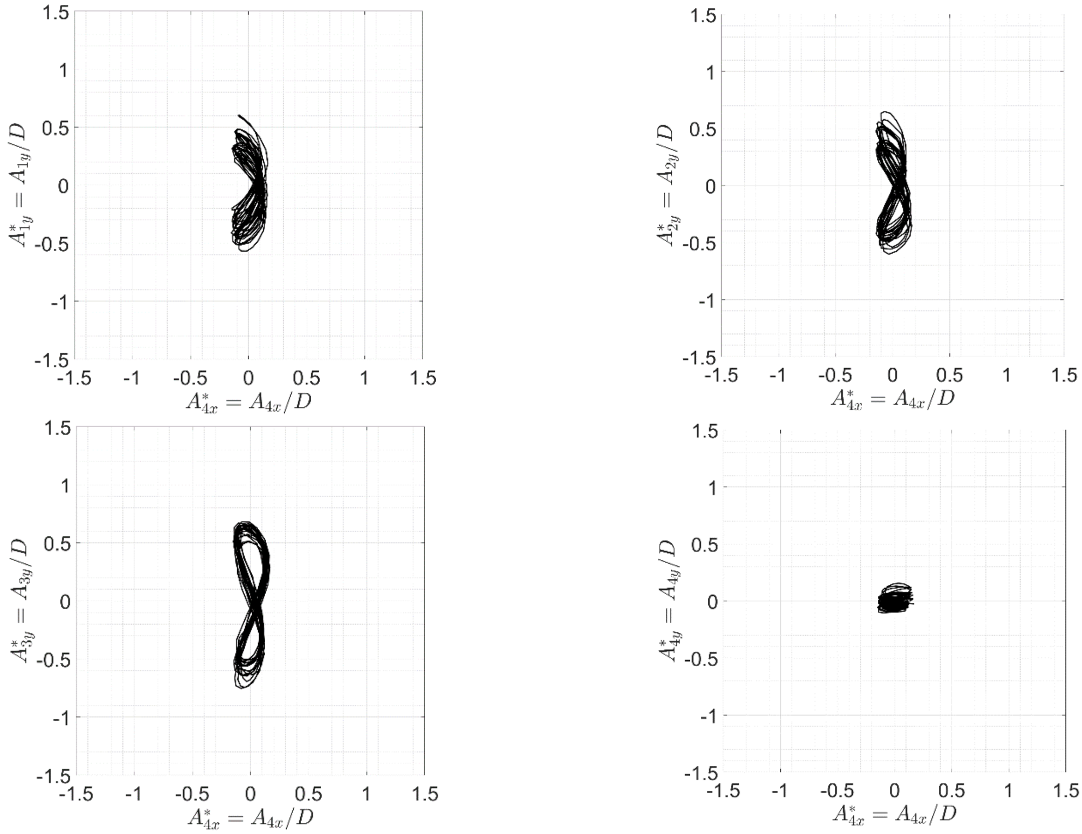


Figure 18: Modal orbits on the XY plane. Pure towing test at $U = 0.28$ m/s ($U^*_{3y} = 7.32$). A^*_{1y}, A^*_{2y}, A^*_{3y} and A^*_{4y} vs. A^*_{4x}.

3.2. VIV Similarities

Figure 19 compares the modal amplitudes and frequency responses exhibited by the experimental models, focusing on the respective modes of interest. The crosswise modal amplitudes demonstrate excellent agreement between them. The three initial branches occur at the same reduced velocity region and exhibit nearly identical amplitude peaks. Similarly, the amplitude jumps from the initial to the upper branches occur at the same velocity. The reduced velocity region in which the upper branches occur corresponds very well. Models BH-1 and BH-3 exhibit similar amplitudes and lack a distinctive lower branch. Instead, they display a continuous decrease in oscillation up to the highest velocities tested. The discrepancies observed can be attributed to the response of the BH-2 model, which exhibits higher amplitudes at the upper branch and a distinct second amplitude jump, resulting in an amplitude decrease to a lower branch of response. The respective crosswise frequencies of the three models are constrained within a similar and narrow frequency band that varies in a practically continuous and linear manner over the entire test range. In the streamwise direction, the amplitude responses differ in magnitude and also at the peak regions along the test range. The respective frequencies displayed a scattered arrangement.

In Figure 20, a comparison of the respective average drag coefficients is presented. It was calculated as $C_D = 2F_x/\rho_w D L_{ef} U^2$, where F_x is the mean streamwise force, ρ_w is the water density, D is the external diameter, L_{ef} is the effective length (the length projection onto the vertical axis), and U is the towing speed. It is important to note that the drag coefficients vary along the length of the models due to variations in amplitude and inclination to the vertical axis. Accordingly, the resulting coefficients are regarded as representative averages. The obtained values are markedly larger than the typical value of 1.2 documented in the literature for fixed rigid cylinders. This discrepancy is attributed to the effect of the VIV amplitudes, as also has been reported in numerous previous studies. The maximum drag coefficient observed in model BH-1 is approximately $C_D \approx 2$ at $U^*_{1y} = 6$, while models BH-2 and BH-3 exhibit similar maximum coefficients of approximately

$C_D \approx 2.6$, however at distinct reduced velocities, $U^*_{2y} = 7$ and $U^*_{3y} = 4$, respectively. The identification of the reduced velocity range within which the maximum drag coefficients occur is contingent upon the interaction between all the natural modes involved in the dynamics. It can thus be concluded that there may not necessarily be a direct correlation between the main crosswise branch of response (the upper branch) and the aforementioned coefficients, at least for structures exhibiting high flexibility. This is particularly evident in the case of the model BH-3, where the highest values of the coefficients are located in the region of its second crosswise mode resonance, rather than the third mode, which was designed to be the dominant mode and presented the most extended resonance region. In contrast, models BH-1 and BH-2 display maximum coefficients in reduced velocities that are consistent with their upper branch regions.

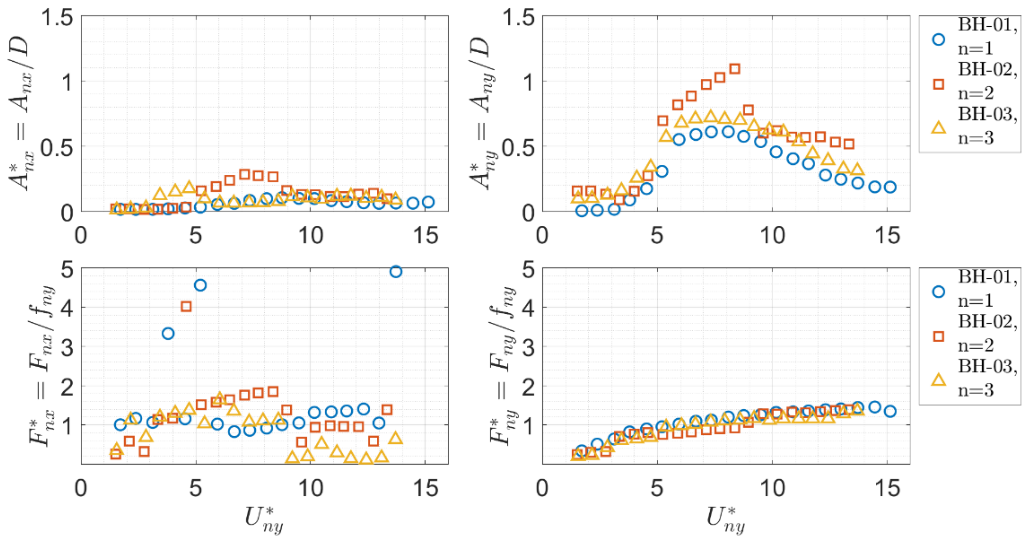


Figure 19: Modal amplitudes and the respective dominant frequencies as function of the reduced velocity U^*_{ny}.

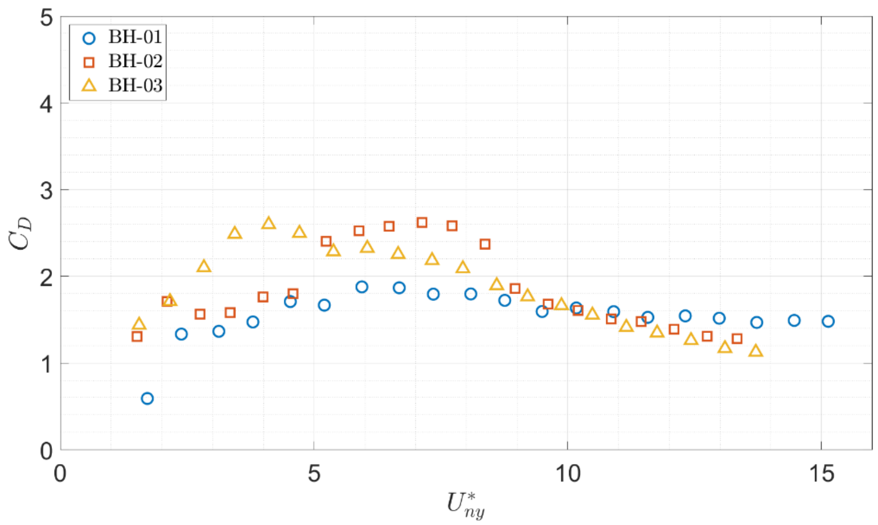


Figure 20: Average drag coefficients as function of the reduced velocity U^*_{ny}.

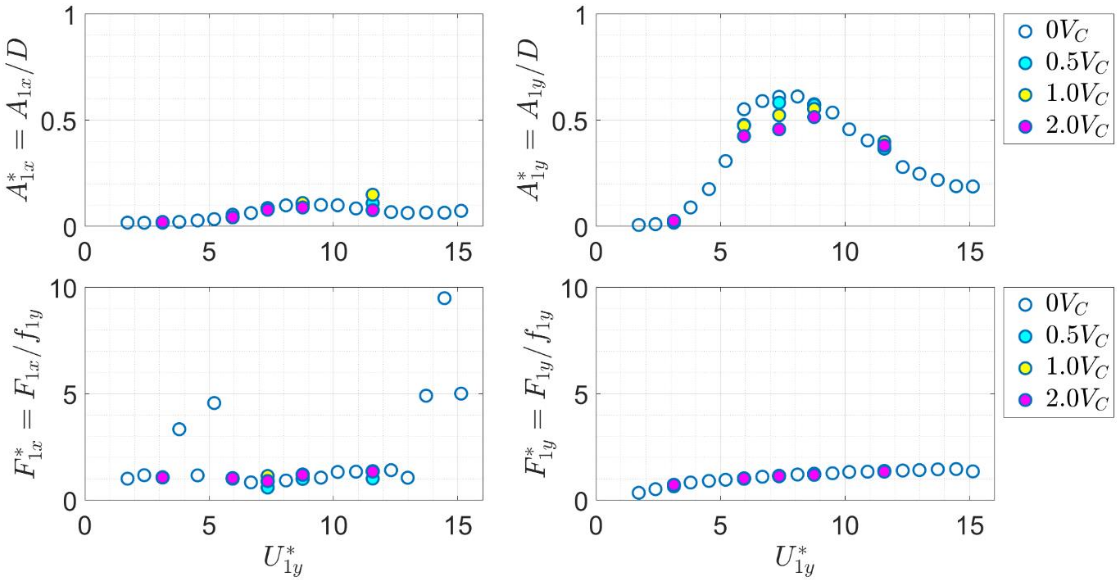
3.3. VIV and Aspirating Flow

This subsection focuses on the tests in which both external and internal flows are simultaneously present. In other words, the focus is now turned on to the effect of the internal aspirating flow on the VIV phenomenon. Would it be mitigated or amplified by the aspirating flow? The analysis will concentrate on the natural mode of interest of each model and its counterpart in the streamwise direction. Five points of external flow velocity were measured with the presence of aspirating flow. The empty markers are related to the case of pure external flow, while solid markers indicate the presence of aspirating flow with the velocity specified in the legend, which corresponds to 0.5, 1.0 and 2.0 times the critical value for instability. Figure 21(a) presents the comparison for the BH-1 model, while the comparison for models BH-2 and BH-3 are presented in Figure 21(b) and Figure 21(c), respectively.

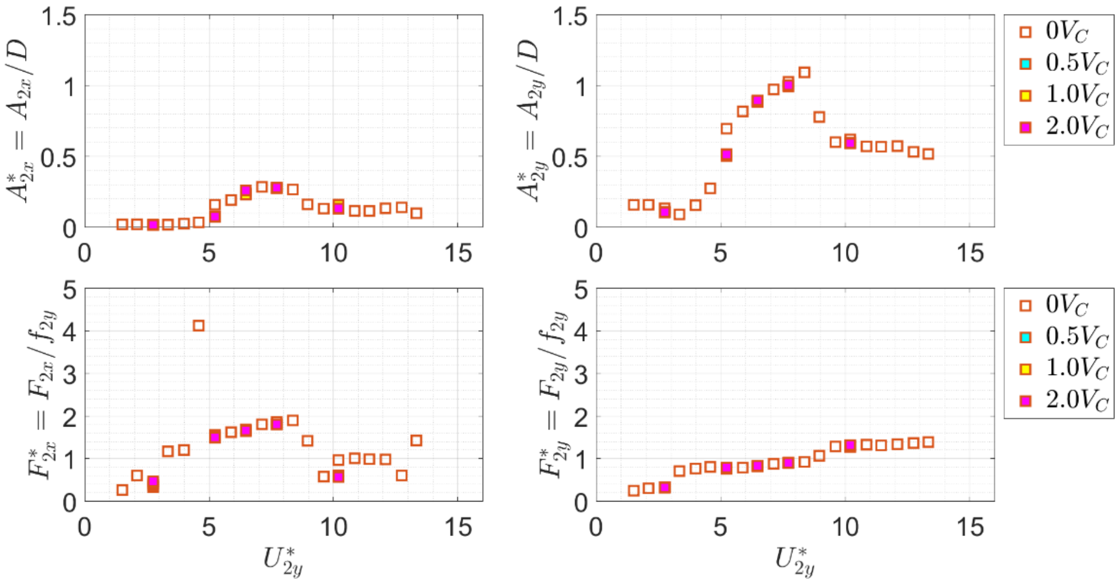
The comparison for the model BH-1 demonstrates a sensible reduction in the crosswise amplitudes, particularly evident in the measurements taken at the upper branch of response. The aspirating flow velocity that exerts the most influence is the $2.0\,V_C$ case. In contrast, models BH-2 and BH-3 exhibit VIV amplitudes that are essentially unaffected by the internal flow regardless of the aspirating velocity. However, a slight decrease in amplitude is observed at the transition from the initial to the upper branch. The respective crosswise frequencies for all models demonstrate minimal variation in magnitude and frequency band constriction. A similar observation can be made with regard to the streamwise frequencies. The BH-1 model's response was the most affected by the aspirating flow, likely because its first natural mode was designed to be the predominant one excited by VIV. Additionally, the unstable mode induced by the aspirating flow also corresponds to the first natural mode, making it the most impacted. In contrast, for the BH-2 and BH-3 models, the predominant natural modes excited by VIV were the second and third modes, respectively, while the instability caused by the aspirating flow also appeared at their first natural modes. As a result, their VIV responses were less affected by the aspirating flow.

It is evident that the VIV phenomenon is predominant over the aspirating effects, regardless of the aspirating flow velocity. No qualitative change in the oscillations caused by the external flow was observed. However, it can be pointed out that the aspirating flow slightly mitigated the VIV oscillations, representing a subtle damping effect, at least in the scale of the present experiment. Therefore, it can be concluded that the VIV is considered the dominant phenomenon over the effects of the aspirating flow.

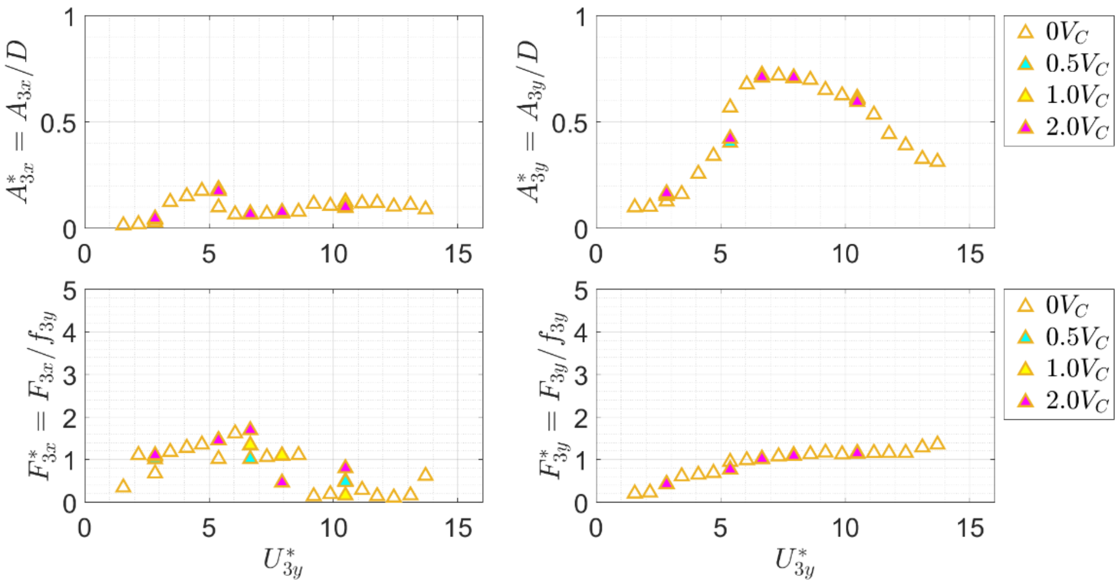
An overall comparison can be made with the numerical study presented by Orsino ([30]), which simulated a cantilevered flexible pipe simultaneously subjected to the effects of internal flow in aspirating condition (at both pre-critical and post-critical velocities) and external flow, represented by a wake-oscillator phenomenological model. The numerical results showed a slight increase in the crosswise VIV amplitudes across the entire reduced velocity range, regardless of the aspirating flow velocity condition. However, the presence of aspirating flow did not qualitatively affect the VIV responses, which continued to exhibit the same response branches in their respective reduced velocity regions as in the case without aspirating flow. Additionally, Finoteli *et al*. (2022, [15]) presented a preliminary numerical-experimental study that directly compares the experimental results of the BH-2 model with numerical results obtained using the in-house model. The numerical model was defined as a cantilevered flexible pipe with the same dimensions and physical properties as the actual experimental model. The dynamic excitation scenarios considered included pure VIV and the combined effects of VIV and aspirating flow in post-critical dynamic regime, at twice the value of the respective critical velocity. The numerical results showed similar VIV amplitude mitigation with minimal change in the dominant frequencies in the presence of aspirating flow, aligning with the experimental results.



a) BH-1 model.



b) BH-2 model.



c) BH-3 model.

Figure 21: Modal amplitudes and the respective dominant frequencies as function of the reduced velocity. Left and right columns are for the streamwise and crosswise directions, respectively. Filled markers indicate aspirating flow.

4. Conclusions

The present study addressed an experimental investigation on the dynamics of three ballasted cantilevered flexible pipes, subjected to the scenarios of (i) pure VIV and (ii) combined VIV and aspirating flow. The models were designed and manufactured in accordance with the conceptual characteristics of Seawater Intake Risers. The first model, BH-1, was excited by VIV in the first natural mode of vibration whereas VIV induced the lock-in with the second and third natural modes in models BH-2 and BH-3, respectively. For all models, the target frequency for VIV excitation was set as 1 Hz. The three models were tested concomitantly in a towing tank, in a side-by-side arrangement without mutual hydrodynamic interferences, to ensure that the external flow velocity was identical for all. An optical system was employed to record the displacement of the models by tracking reflective targets that were affixed along their lengths. The data analysis methodology yielded dynamic responses along the length of the models and for each natural mode of vibration, for each velocity tested.

In model BH-1, the first crosswise natural mode was identified as the dominant, while the second and third natural modes were identified as the dominant ones in models BH-2 and BH-3, respectively. As a consequence of the bending flexibility, the amplitudes of vibration are constituted by contributions from multiple natural modes, which vary in both magnitude and frequency.

Noteworthy findings were observed in the interactions between the natural modes involved in the dynamics. Initially, a succession of upper branches was identified, from lower to higher natural modes, following the increase in the flow velocity, specifically in models BH-2 and BH-3. This was a consequence of a larger set of natural modes being likely to be excited. In contrast to elastically mounted rigid cylinders, flexible ones can exhibit this distinctive behavior due to the range of flow velocities under investigation, which may cause vortex shedding in harmonics compatible with more than one natural frequency of the system. The classical dynamics demonstrated by an elastically mounted rigid cylinder, comprising an initial-upper-lower branches sequence, is transformed into a combination of those, with one branch sequence designated for each mode that may be excited. A practical limitation to this effect, particularly in the context of cantilevered cylinders, is the inclination to the vertical axis caused by drag, which reduces the projection of the cylinder’s length to the incoming flow and also results in axial components of the flow that affect the structure of vortex shedding.

A second remarkable feature observed in the experiments was the internal resonance effect, whereby a transfer of energy occurs from a natural mode in its upper branch of response to other modes. This phenomenon, also characteristic of flexible systems, can be attributed to structural and hydrodynamic factors. As a specific crosswise natural mode exhibits its main resonance (the upper branch of response), a transfer of energy occurs, stimulating other natural modes within the same frequency band exhibited by the resonant mode. Accordingly, the mode exhibiting the upper branch is regarded as the dominant one, while the excited modes are considered the secondary. Additionally, it was also observed that natural modes can be excited at higher reduced velocities than those from their main resonance region. Furthermore, the excitation of secondary modes is not confined to parallel modes, i.e., modes in the same plane. The findings have revealed that streamwise modes were also excited by crosswise dominant modes, indicating that the internal resonance may act on an orthogonally as well.

Dual resonance regimes were identified primarily in models BH-2 and BH-3. This regime is defined by a frequency ratio of 2 between two orthogonal modes, indicating a more energetic region of oscillation. This phenomenon was identified to occur concomitantly between a variety of natural modes. As a matter of fact, the secondary modes may experience the same dual resonance regime as the dominant one. Additionally to the in-line dominant peak of response exhibiting a 2:1 frequency ratio relative to the dominant peak in the crosswise direction, the in-line response also presented weaker harmonics matching the respective frequencies of other natural modes, despite the presence of large deflections caused by drag effects.

Lastly, the comparison between the scenarios of pure external flow and external flow with aspirating flow clearly demonstrated that the VIV phenomenon remained dominant, with minimal influence from the aspirating effects. Although the aspirating flow produced a weak damping effect by slightly reducing the VIV crosswise amplitudes, it did not affect the corresponding crosswise frequencies, which maintained the same values and exhibited similarly narrow bands. No qualitative difference was observed when compared to the pure VIV scenario, i.e., the amplitude branches remained in the same range of reduced velocity with practically the same amplitudes, corresponding frequencies, and jumps, regardless of the ratio of aspirating critical velocity imposed, reinforcing the limited capacity of the aspirating flow to interfere with the VIV.

This study reinforces the complex dynamics presented by cantilevered flexible pipes under the effects of external flow and adds new findings on the effects of aspirating flow. The presented findings may serve as new benchmarks for the dynamics of cantilevered flexible pipes subjected to pure external flow excitation as well as combined external and aspirating flows excitation. This work is intended to serve as a reference for the calibration and simulation of analytical and numerical models of similar systems.

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Data Availability

The raw data recorded in the experiments are available from the corresponding author on a reasonable request.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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Appendix A. Material characterization of the flexible hoses

This appendix presents a description of the dynamic characterization of the rubber hose material utilized as flexible pipes for manufacturing the experimental models. As previously stated in Section 2, the hose parameters were employed to define the length of the models using the in-house code. However, given that the VIV tests subject the material to a dynamic excitation, utilizing a static bending stiffness would yield inaccurate length estimation. Moreover, the hose's stiffness is provided by the combination and interaction of the rubber and fiber layer resistances (see Fig. A2). Consequently, a measured dynamic bending stiffness was essential.

In order to address this need, a dynamic characterization test was conducted at the Centro de Caracterização e Desenvolvimento de Materias (CCDM) in the city of São Carlos, employing cyclic tensile tests. Each test commenced with the application of a pre-load of 50 N, which was then followed by six tension-relaxation cycles. Throughout the duration of the test, load cells and optical tracking were employed to measure the loads and strains, respectively. The tests were executed at the excitation frequencies of 0.5 Hz, 0.75 Hz and 1.0 Hz. No value above the last frequency was possible due to the limitations of the hardware.

The resulting "force x strain" curves display an elliptical shape for each cycle, from which the dynamic axial stiffness was calculated. This calculation employs the angle of inclination formed by the diagonal lines connecting the maximum and minimum values of each ellipse. The structural damping coefficient was calculated from the area of each ellipse, though it was not employed in the models' dimensioning. Tables A1 and A2 present the measured dynamic axial stiffness and the structural damping, respectively. The corresponding average values and standard deviation are also presented. Figure A 1 displays an example of the "force x strain" curve obtained during one of the 1 Hz tests, along with two plots presenting the corresponding results.

As the models were designed with the natural frequencies of their crosswise modes of interest situated in close proximity to 1.0 Hz, the results obtained from the cyclic test at this frequency were utilized as reference. Accordingly, the measured dynamic axial stiffness (at 1.0 Hz cyclic test) is $EA_H = 12.5 \cdot 10^3$ N. Under the assumption of constant and equal to the nominal internal and external diameters along the length of the hose, the dynamic Young's modulus was calculated to be $E_H = 26.4 \cdot 10^6$ N/m^2. From the inferred Young's modulus, the obtained dynamic bending stiffness was $EI_H = 1.23$ Nm^2. It can be seen from Tab. 1 that the calculated and measured natural frequencies differ only slightly, indicating that the approach used to obtain the dynamic bending stiffness was reasonable.

In regard to the ballast design, meticulous attention was paid to circumvent any imperfections or discontinuities at the hose-ballast interface. This was achieved through the implementation of a specially crafted thread interference. To permit internal flow, the anchors were constructed from a 50 mm length of stainless-steel pipe, introduced and clamped within the hose's upper end. This piece of pipe was designed with a sufficient length to be clamped at structural bars and connected to the pumping system at the testing site (for further details, please refer to Figs. A2 and A3).

Table A 1: Measured dynamic axial stiffness for the hose samples.

Sample	Tests	Pre-load [N]	Cycles	0.5 Hz EA [N]	0.5 Hz Standard Deviation [N]	0.5 Hz Standard Deviation [%]	0.75 Hz EA [N]	0.75 Hz Standard Deviation [N]	0.75 Hz Standard Deviation [%]	1 Hz EA [N]	1 Hz Standard Deviation [N]	1 Hz Standard Deviation [%]
1	6	50	6	11121,0	141,1	1,27%	11421,0	79,5	0,70%	11930,0	157,79	1,32%
2	6	50	6	11557,0	130,8	1,13%	11931,0	135,5	1,14%	12446,0	198,87	1,60%
3	6	50	6	12273,0	265,8	2,17%	12156,0	455,0	3,74%	13227,0	300,47	2,27%
Average value				**11650,3**	**179,22**	**1,52%**	**11836,0**	**223,31**	**1,86%**	**12534,3**	**219,04**	**1,73%**

Table A 2: Measured structural damping coefficient for the hose samples.

Sample	Tests	Pre-load [N]	Cycles	0.5 Hz C	0.5 Hz Standard Deviation	0.5 Hz Standard Deviation [%]	0.75 Hz C	0.75 Hz Standard Deviation	0.75 Hz Standard Deviation [%]	1 Hz C	1 Hz Standard Deviation	1 Hz Standard Deviation [%]
1	6	50	6	9,35E-05	3,77E-06	4,03%	5,06E-05	1,84E-06	3,63%	4,73E-05	1,42E-06	3,01%
2	6	50	6	8,44E-05	1,68E-06	1,99%	4,74E-05	1,92E-06	4,06%	3,62E-05	2,98E-06	8,22%
3	6	50	6	7,61E-05	1,95E-06	2,57%	4,49E-05	1,57E-06	3,49%	4,24E-05	3,76E-06	8,87%
Average value				**8,47E-05**	**2,47E-06**	**2,86%**	**4,76E-05**	**1,78E-06**	**3,72%**	**4,20E-05**	**2,72E-06**	**6,70%**

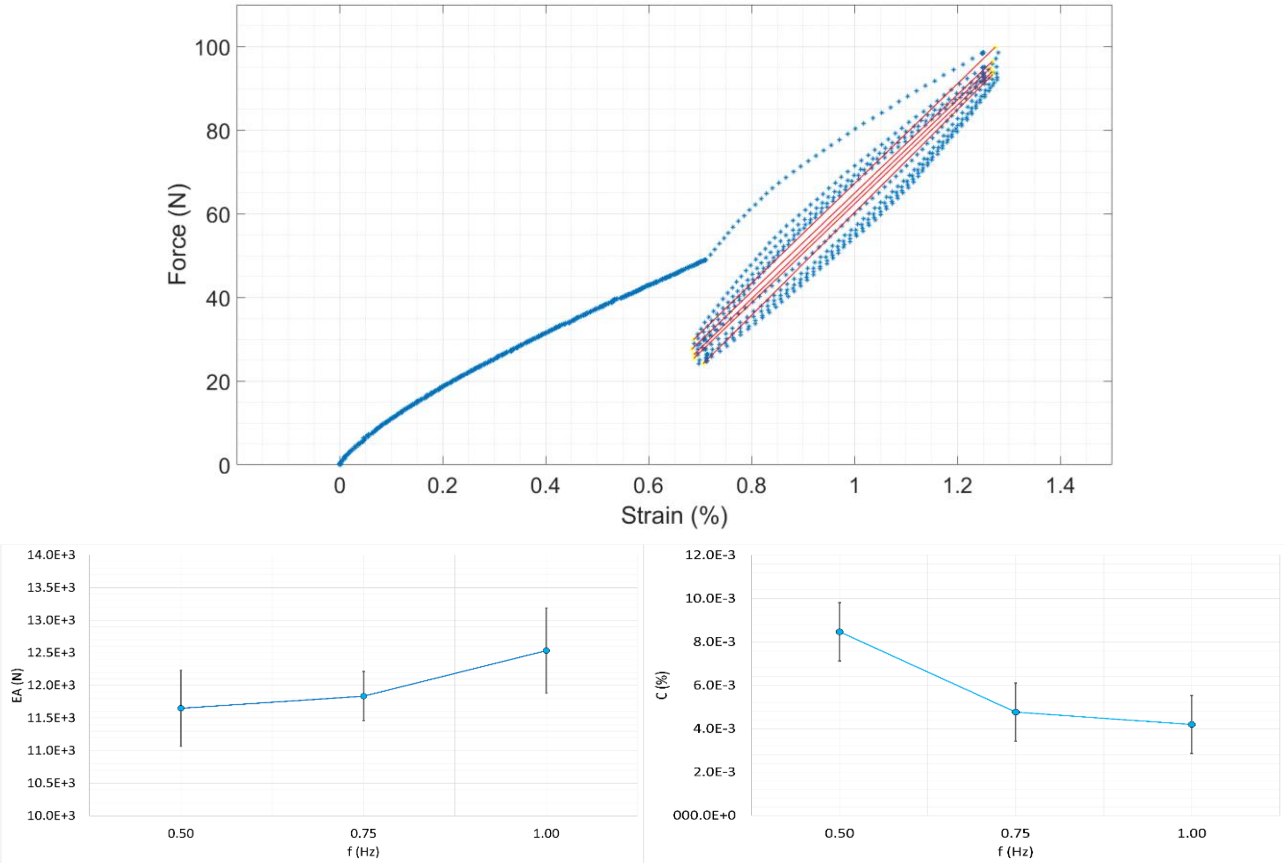

Figure A 1: (Above) Example of the dynamic tensile test curve at 1 Hz frequency. In blue: Force [N] vs Strain [%]; in red: diagonals of the ellipses. (Below on the left) Average dynamic axial stiffness and (Below on the right) average structural damping coefficient, both as functions of the frequency of excitation. Vertical bars indicate the standard deviation.

Figure A 2: Cross section view of the rubber hose and detail of the hose-ballast connection based on thread interference.

Figure A 3: The three experimental models ready for tests. (Above) Comparison of length scale. (Below) Detail of the reflective tapes and anchorage systems.